

# 4.4 Image Sharpening

- Introduction
- First-order derivative
- Second-order derivative
- High-pass filter in frequency domain

# 4.4 Image Sharpening

## 4.4.1 Introduction

**Purposes:** easy to detect edges  
highlight fine detail

**Request:** insensitive to noise

# 4.4 Image Sharpening

## 4.4.1 Introduction: Sharpening filters

- (1) First-order derivative : Robert  
Prewitt Sobel  
Robinson Kirsch
- (2) Second-order derivative : Laplacian  
LoG
- (3) High-pass filter in frequency domain :  
Idea high-pass filter (IHPF)  
Butterworth high-pass filter (BHPF)  
Gaussian high-pass filter (GHPF)

# 4.4 Image Sharpening

## 4.4.2 first-order derivative : foundation

The derivatives of a digital function are defined in terms of differences

For first derivative it must be

- (1) zero in flat segment
- (2) nonzero at the onset of a gray-level step or ramp
- (3) nonzero along ramp

For second derivative it must be

- (1) zero in flat segment
- (2) nonzero at the onset and end of a gray-level step or ramp
- (3) zero along ramp of constant slope

## 4.4 Image Sharpening

### 4.4.2 first-order derivative : foundation

A basic definition of the first-order derivative of a 1-D function  $f(x)$  is the difference

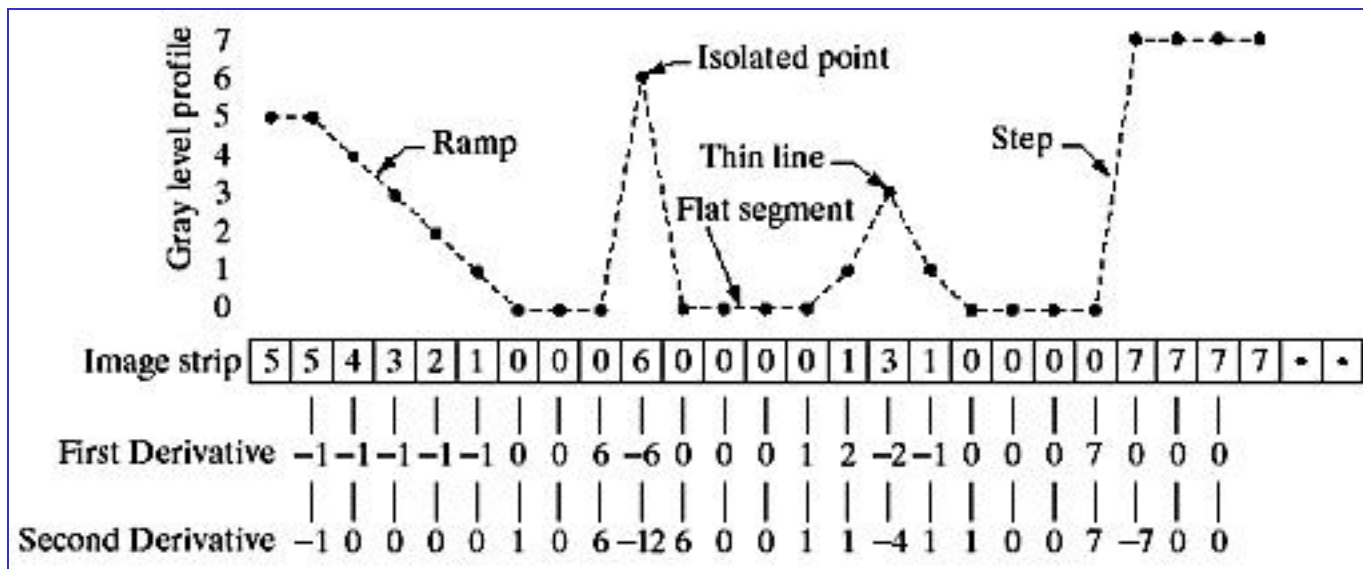
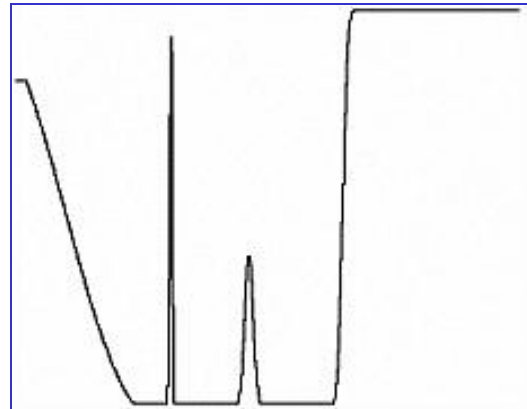
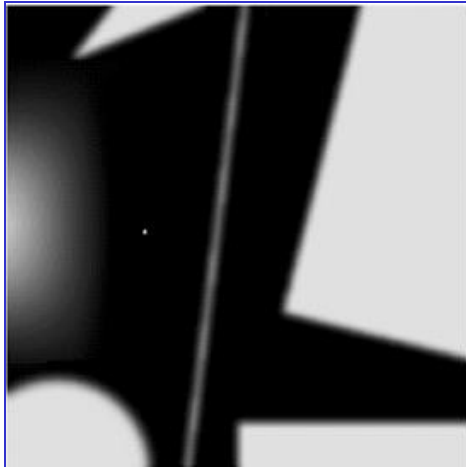
$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

Similarly, we define a second-order derivative as the difference

$$\begin{aligned}\frac{\partial^2 f}{\partial x^2} &= [f(x+1) - f(x)] - [f(x) - f(x-1)] \\ &= f(x+1) + f(x-1) - 2f(x)\end{aligned}$$

# 4.4 Image Sharpening

## 4.4.2 first-order derivative : foundation



# 4.4 Image Sharpening

## 4.4.2 first-order derivative : foundation

First-order derivatives of a digital image are based on various approximations of the 2-D **gradient**. It is defined as the **vector**:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix}^T$$

Magnitude:

$$|\nabla f| = \left[ \left( \frac{\partial f}{\partial x} \right)^2 + \left( \frac{\partial f}{\partial y} \right)^2 \right]^{\frac{1}{2}} \longrightarrow |\nabla f| \approx \left| \left( \frac{\partial f}{\partial x} \right) \right| + \left| \left( \frac{\partial f}{\partial y} \right) \right|$$

Direction:

$$\alpha(x, y) = \tan^{-1} \left( \frac{G_y}{G_x} \right)$$

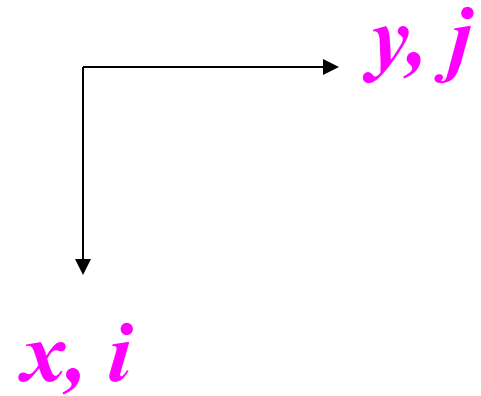
$$|\nabla f| \approx \max \left\{ \left| \frac{\partial f}{\partial x} \right|, \left| \frac{\partial f}{\partial y} \right| \right\}$$

## 4.4 Image Sharpening

### 4.4.2 first-order derivative : foundation

where  $G_x = \frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$

$$G_y = \frac{\partial f}{\partial y} = f(x, y+1) - f(x, y)$$



These equations can be implemented using masks

$$G_x = \begin{vmatrix} -1 & 0 \\ 1 & 0 \end{vmatrix}$$

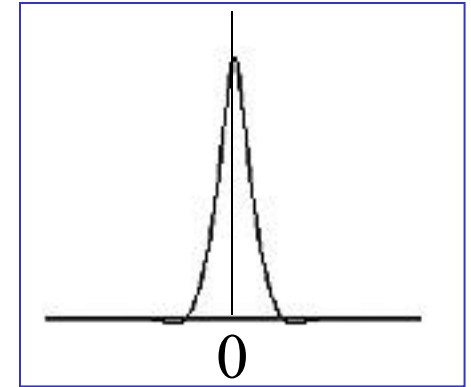
$$G_y = \begin{vmatrix} -1 & 1 \\ 0 & 0 \end{vmatrix}$$



## 4.4 Image Sharpening

### 4.4.2 first-order derivative : display

$$f(x, y) \quad \longrightarrow \quad g(x, y) = \nabla f$$



$$(1) \quad g(x, y) = |\nabla f|$$

$$(2) \quad g(x, y) = f(x, y) + |\nabla f|$$

$$(3) \quad \text{Linear scaling} \quad g(x, y)$$

$$(4) \quad g(x, y) = \begin{cases} L_b & |\nabla f| < T \\ L_t & \text{otherwise} \end{cases}$$

# 4.4 Image Sharpening

## 4.4.2 first-order derivative : foundation

|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 10.10**

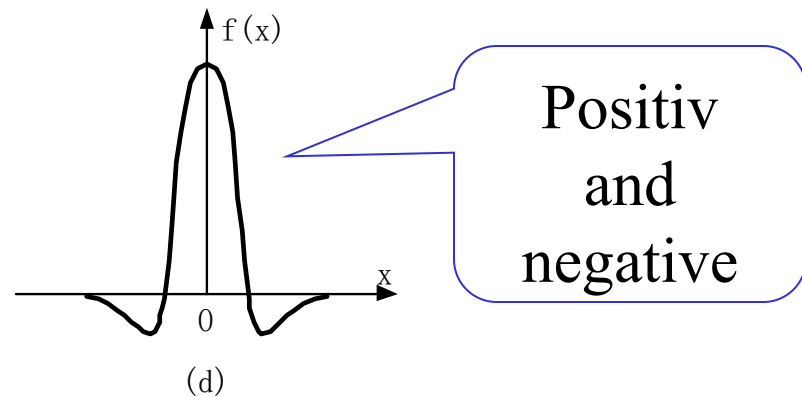
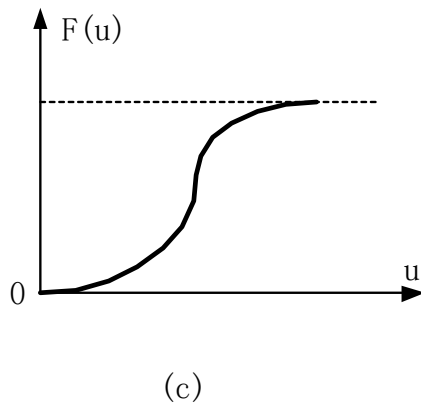
(a) Original image. (b)  $|G_x|$ , component of the gradient in the  $x$ -direction. (c)  $|G_y|$ , component in the  $y$ -direction. (d) Gradient image,  $|G_x| + |G_y|$ .



# 4.4 Image Sharpening

## 4.4.2 first-order derivative : foundation

**Spatial filter:** the sign of filter coefficients



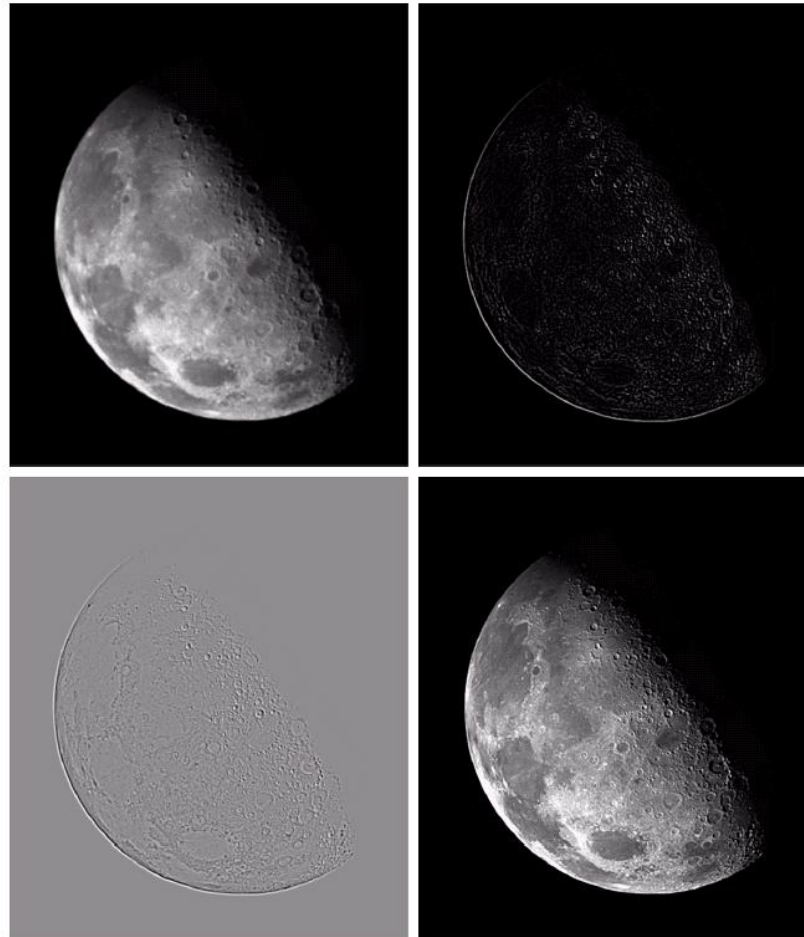
# 4.4 Image Sharpening

## 4.4.2 first-order derivative : display

a b  
c d

**FIGURE 3.40**

(a) Image of the North Pole of the moon.  
(b) Laplacian-filtered image.  
(c) Laplacian image scaled for display purposes.  
(d) Image enhanced by using Eq. (3.7-5).  
(Original image courtesy of NASA.)



# 4.4 Image Sharpening

## 4.4.2 first-order derivative : gradient operators

Roberts operator

$$G_x = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} \quad G_y = \begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix}$$

Prewitt operator

$$G_x = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{vmatrix} \quad G_y = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{vmatrix}$$

Sobel operator

$$G_x = \begin{vmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{vmatrix} \quad G_y = \begin{vmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{vmatrix}$$

## 4.4 Image Sharpening

### 4.4.2 first-order derivative : gradient operators

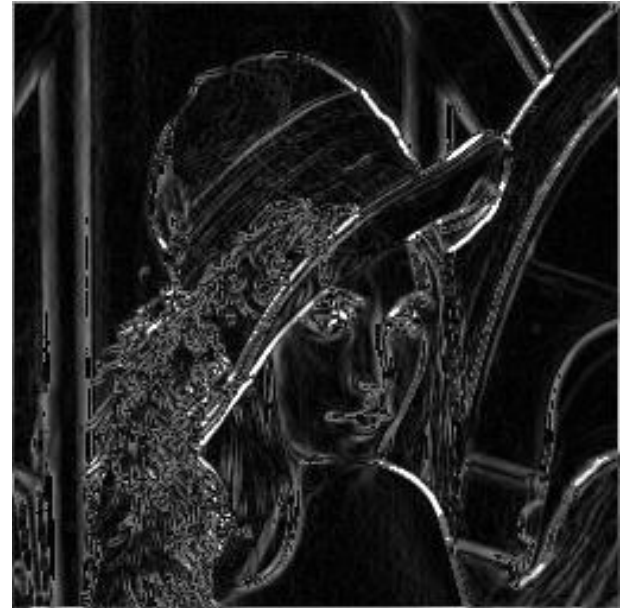
#### Roberts operator



# 4.4 Image Sharpening

## 4.4.2 first-order derivative : gradient operators

### Prewitt operator



# 4.4 Image Sharpening

## 4.4.2 first-order derivative : gradient operators

### Sobel operator

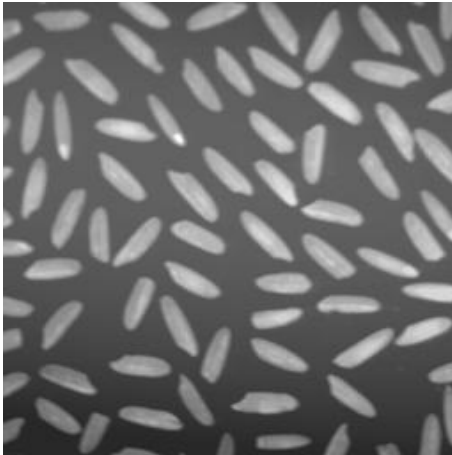




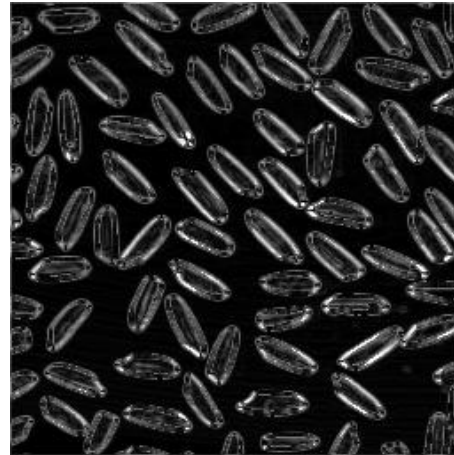
# 4.4 Image Sharpening

## 4.4.2 first-order derivative : gradient operators

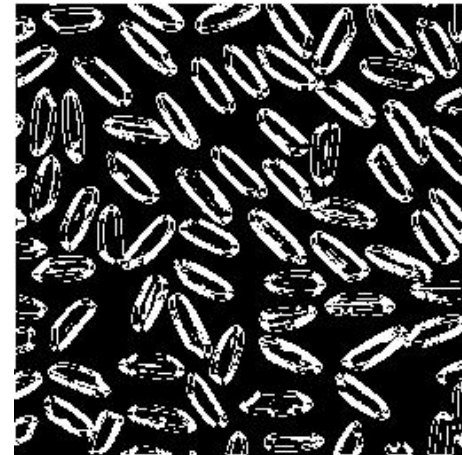
### Sobel operator



original



Sobel



Binary display

## 4.4 Image Sharpening

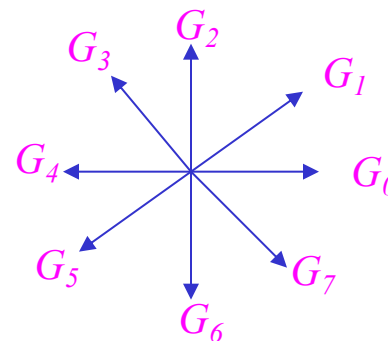
### 4.4.2 first-order derivative : gradient operators

#### Robinson

$$G_0 = \begin{vmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{vmatrix} \quad G_1 = \begin{vmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{vmatrix} \quad G_2 = \begin{vmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{vmatrix} \quad G_3 = \begin{vmatrix} -2 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 2 \end{vmatrix}$$

$$G_4 = \begin{vmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{vmatrix} \quad G_5 = \begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \\ -2 & -1 & 0 \end{vmatrix} \quad G_6 = \begin{vmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{vmatrix} \quad G_7 = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & -2 \end{vmatrix}$$

$$|\nabla f| \approx \max_{i=0,1\dots 7} \{G_i\}$$



## 4.4 Image Sharpening

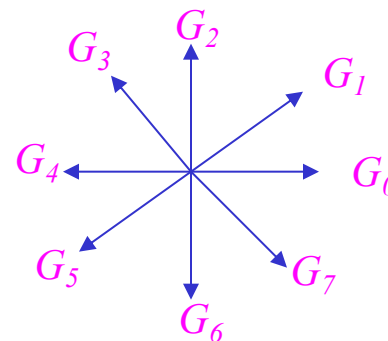
### 4.4.2 first-order derivative : gradient operators

Kirsch

$$G_0 = \begin{vmatrix} 5 & -3 & -3 \\ 5 & 0 & -3 \\ 5 & -3 & -3 \end{vmatrix} \quad G_1 = \begin{vmatrix} -3 & -3 & -3 \\ 5 & 0 & -3 \\ 5 & 5 & -3 \end{vmatrix} \quad G_2 = \begin{vmatrix} -3 & -3 & -3 \\ -3 & 0 & -3 \\ 5 & 5 & 5 \end{vmatrix} \quad G_3 = \begin{vmatrix} -3 & -3 & -3 \\ -3 & 0 & 5 \\ -3 & 5 & 5 \end{vmatrix}$$

$$G_4 = \begin{vmatrix} -3 & -3 & 5 \\ -3 & 0 & 5 \\ -3 & -3 & 5 \end{vmatrix} \quad G_5 = \begin{vmatrix} -3 & 5 & 5 \\ -3 & 0 & 5 \\ -3 & -3 & -3 \end{vmatrix} \quad G_6 = \begin{vmatrix} 5 & 5 & 5 \\ -3 & 0 & -3 \\ -3 & -3 & -3 \end{vmatrix} \quad G_7 = \begin{vmatrix} 5 & 5 & -3 \\ 5 & 0 & -3 \\ -3 & -3 & -3 \end{vmatrix}$$

$$|\nabla f| \approx \max_{i=0,1\dots7} \{G_i\}$$



## 4.4 Image Sharpening

### 4.4.3 second-order derivative : definitions

The second-order derivative is defined as

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

where  $\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

namely  $\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)]$

## 4.4 Image Sharpening

### 4.4.3 second-order derivative : Laplacian

It is well known as *Laplacian* operator. It can be implemented as a mask

$$G = \begin{vmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{vmatrix}$$

## 4.4 Image Sharpening

### 4.4.3 second-order derivative : Laplacian

Experiment result  $g(x, y) = f(x, y) + \nabla^2 f(x, y)$



## 4.4 Image Sharpening

### 4.4.3 second-order derivative : Laplacian

natural result:



## 4.4 Image Sharpening

### 4.4.3 second-order derivative : LoG

Smoothing first

$$h(r) = -e^{-\frac{r^2}{2\sigma^2}}$$

$$r = x^2 + y^2$$

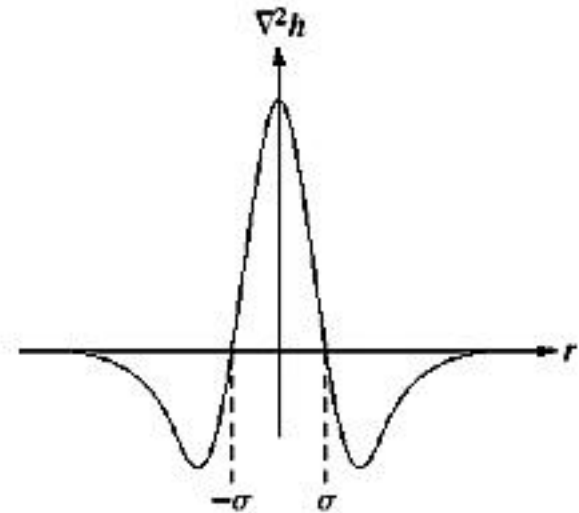
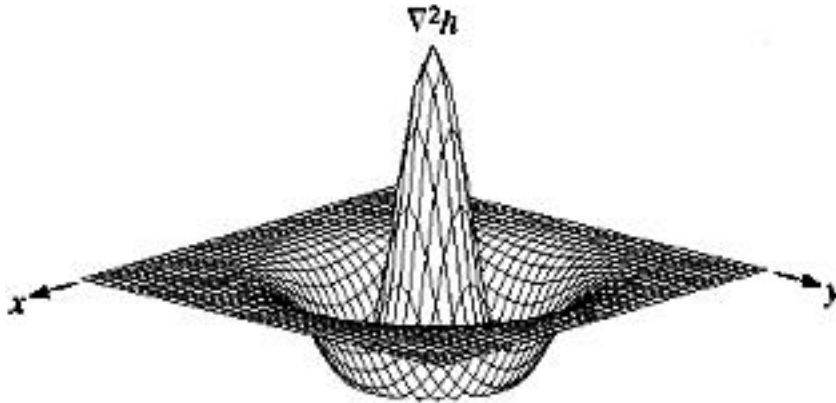
Laplacian sharpening

$$\nabla^2 h(r) = -\left[\frac{r^2 - \sigma^2}{\sigma^4}\right] e^{-\frac{r^2}{2\sigma^2}}$$



# 4.4 Image Sharpening

## 4.4.3 second-order derivative : LoG



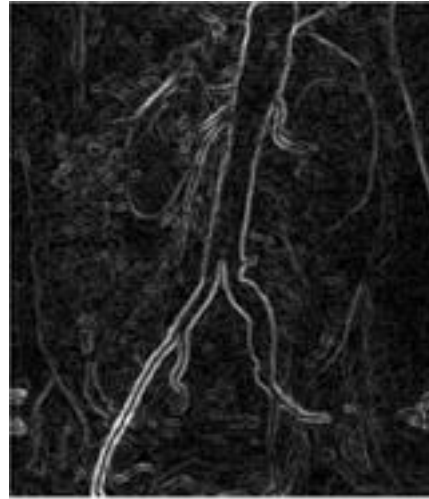
|    |    |    |    |    |
|----|----|----|----|----|
| 0  | 0  | -1 | 0  | 0  |
| 0  | -1 | -2 | -1 | 0  |
| -1 | -2 | 16 | -2 | -1 |
| 0  | -1 | -2 | -1 | 0  |
| 0  | 0  | -1 | 0  | 0  |

## 4.4 Image Sharpening

### 4.4.3 second-order derivative : LoG



original



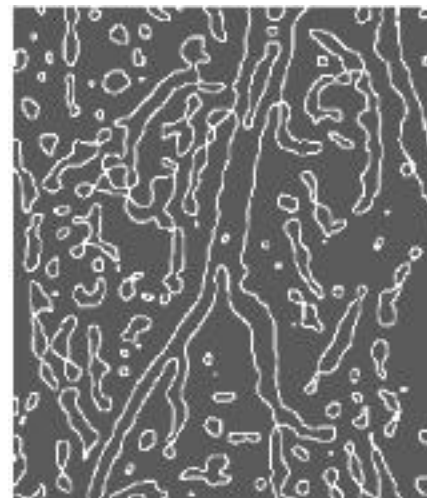
Sobel



LoG



Thresholded LoG



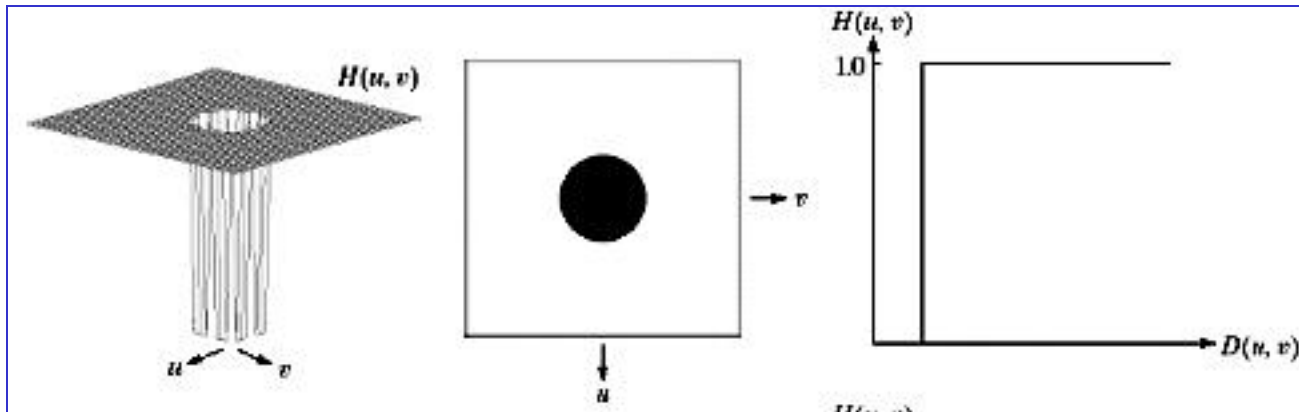
Zero cross

## 4.4 Image Sharpening

### 4.4.3 High-pass filter: Ideal highpass filter (IHPF)

formula

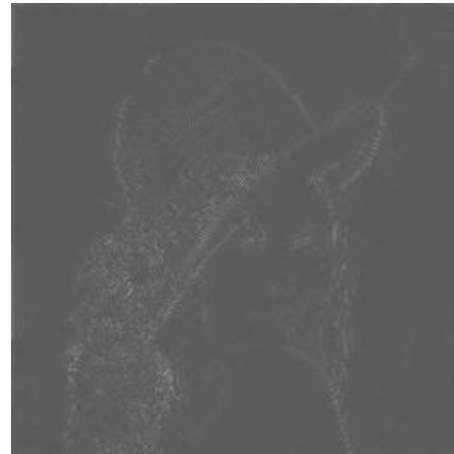
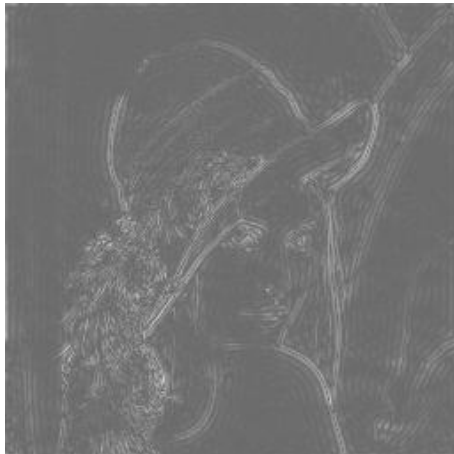
$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$



## 4.4 Image Sharpening

### 4.4.3 High-pass filter: Ideal highpass filter (IHPF)

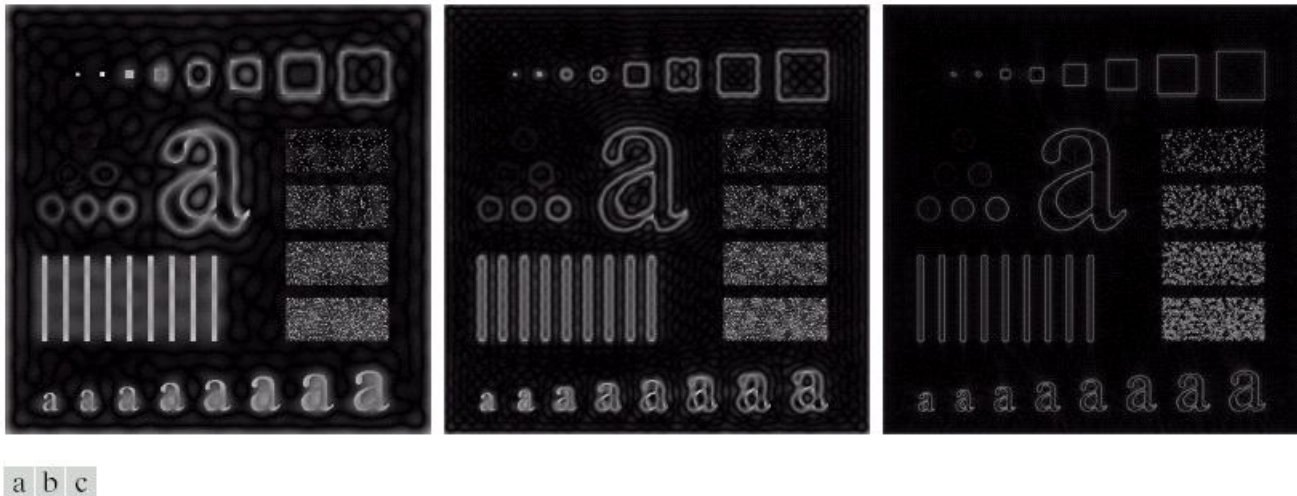
cutoff frequencies set at radii values of 15、30、80



## 4.4 Image Sharpening

### 4.4.3 High-pass filter: Ideal highpass filter (IHPF)

cutoff frequencies set at radii values of 15、30、80



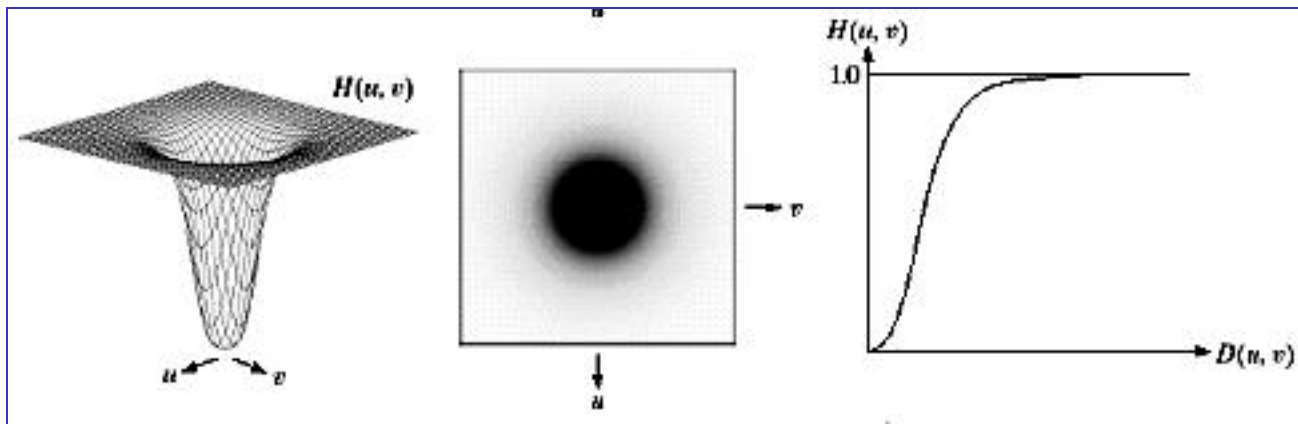
**FIGURE 4.24** Results of ideal highpass filtering the image in Fig. 4.11(a) with  $D_0 = 15, 30$ , and  $80$ , respectively. Problems with ringing are quite evident in (a) and (b).

## 4.4 Image Sharpening

### 4.4.3 High-pass filter: Butterworth highpass filter (BHPF)

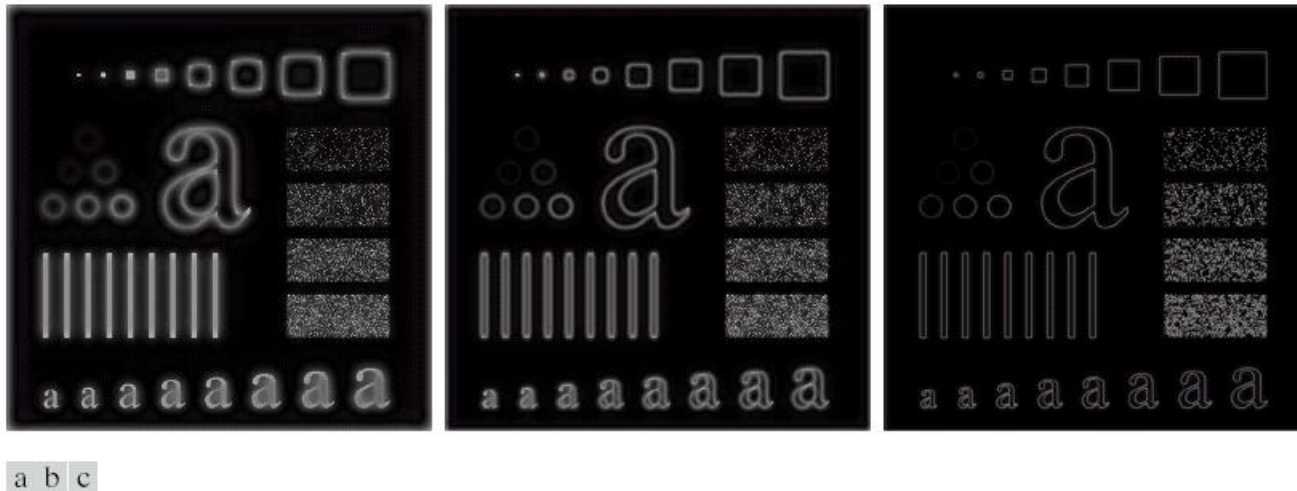
formula

$$H(u, v) = \frac{1}{1 + [D_0 / D(u, v)]^{2n}}$$



## 4.4 Image Sharpening

### 4.4.3 High-pass filter: Butterworth highpass filter (BHPF) cutoff frequencies set at radii values of 15、30、80



**FIGURE 4.25** Results of highpass filtering the image in Fig. 4.11(a) using a BHPF of order 2 with  $D_0 = 15$ , 30, and 80, respectively. These results are much smoother than those obtained with an ILPF.



## 4.4 Image Sharpening

### 4.4.3 High-pass filter: Butterworth highpass filter (BHPF)

cutoff frequencies set at radii values of 15、30、80.



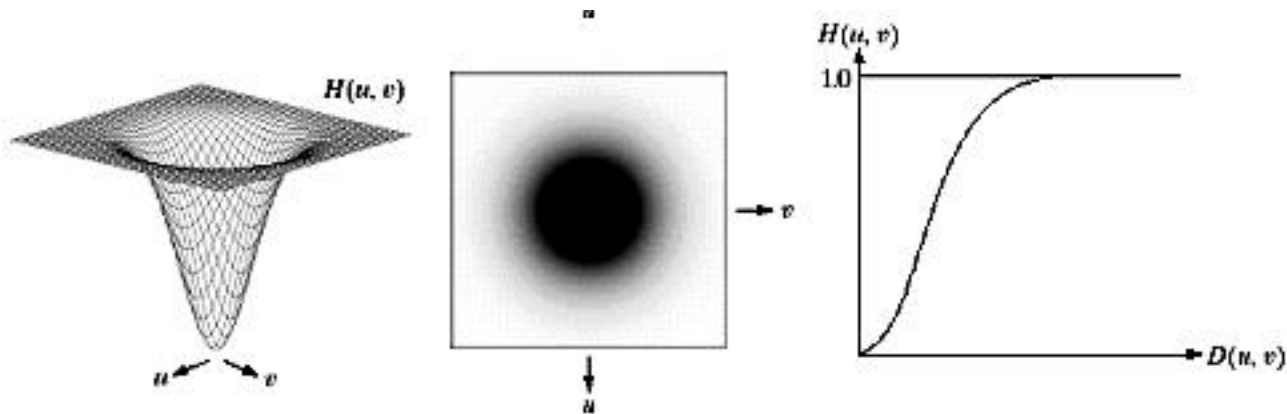


## 4.4 Image Sharpening

### 4.4.3 High-pass filter: Gaussian highpass filter (GHPF)

formula

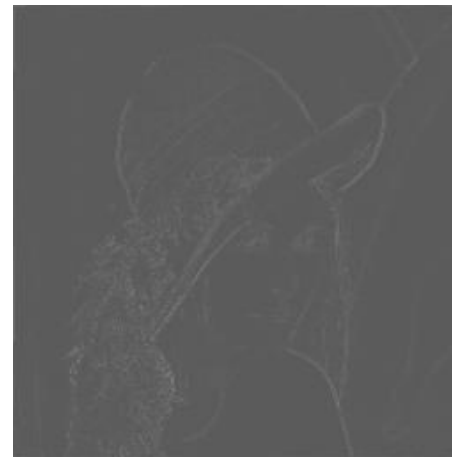
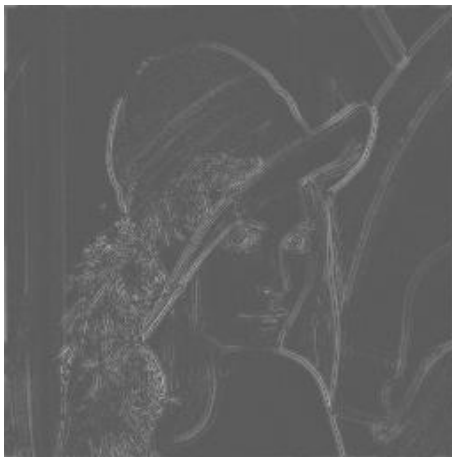
$$H(u, v) = 1 - e^{-D^2(u, v) / 2D_0^2}$$



## 4.4 Image Sharpening

### 4.4.3 High-pass filter: Gaussian highpass filter (GHPF)

cutoff frequencies set at radii values of 15、30、80.



## 4.4 Image Sharpening

### 4.4.3 High-pass filter: Homomorphic filter

An image  $f(x,y)$  can be expressed as the product of illumination and reflectance components

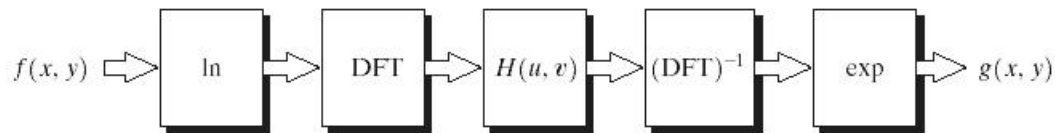
$$f(x, y) = i(x, y)r(x, y)$$

let

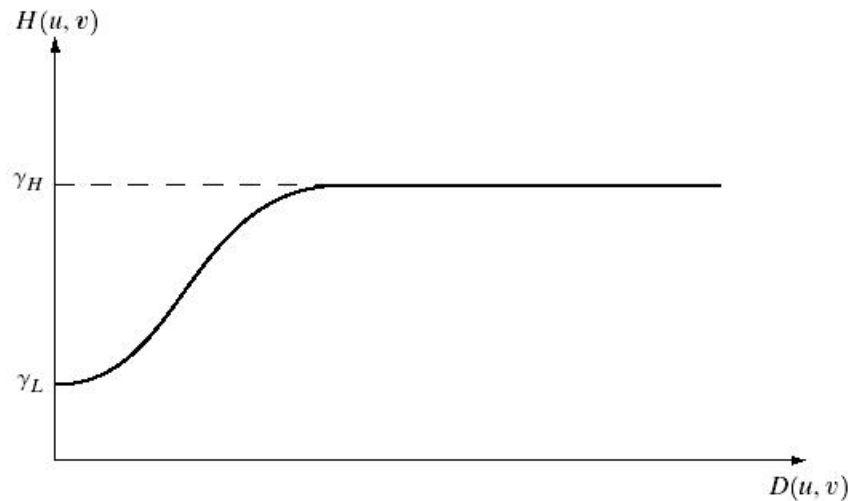
$$z(x, y) = \ln f(x, y) = \ln i(x, y) + \ln r(x, y)$$

# 4.4 Image Sharpening

## 4.4.3 High-pass filter: Homomorphic filter



**FIGURE 4.31**  
Homomorphic  
filtering approach  
for image  
enhancement.



**FIGURE 4.32**  
Cross section of a  
circularly  
symmetric filter  
function.  $D(u, v)$   
is the distance  
from the origin of  
the centered  
transform.

## 4.4 Image Sharpening

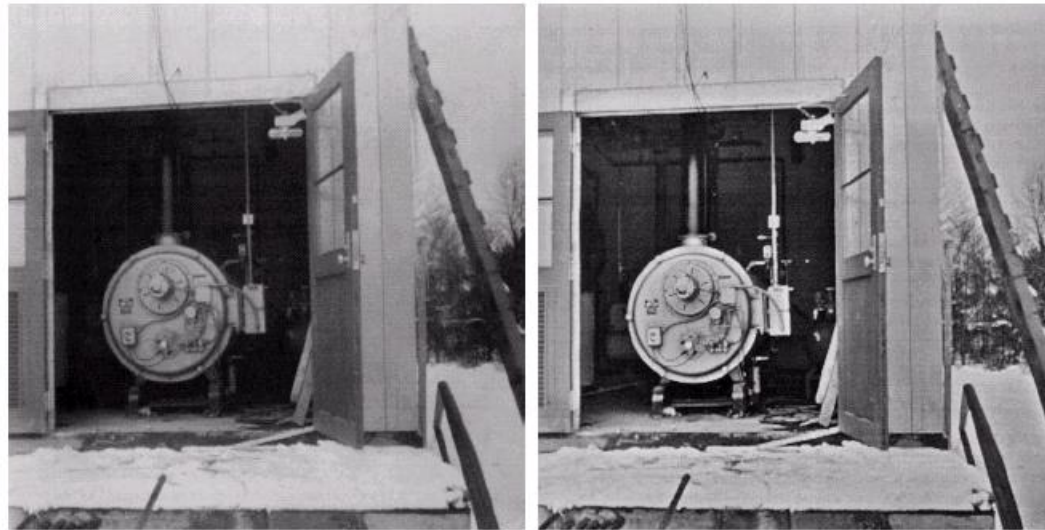
### 4.4.3 High-pass filter: Homomorphic filter

#### Experimental result

a b

**FIGURE 4.33**

(a) Original image. (b) Image processed by homomorphic filtering (note details inside shelter). (Stockham.)

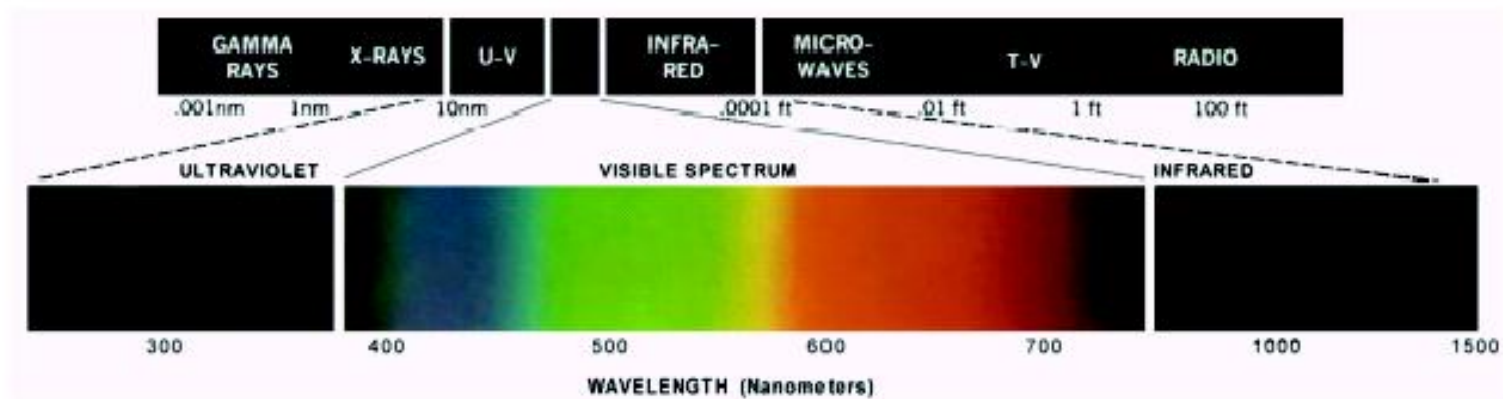


# 4.5 Color Image Enhancement

- Introduction
- Pseudo color image processing
- Full color enhancement
- Noise in color image

# 4.5 Color Image Enhancement

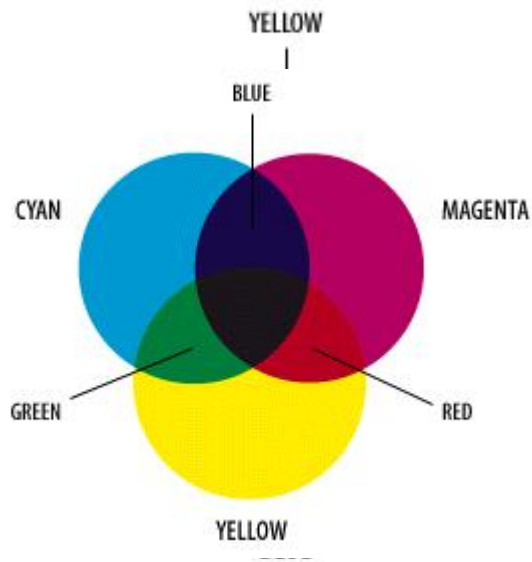
## 4.5.1 Introduction: review



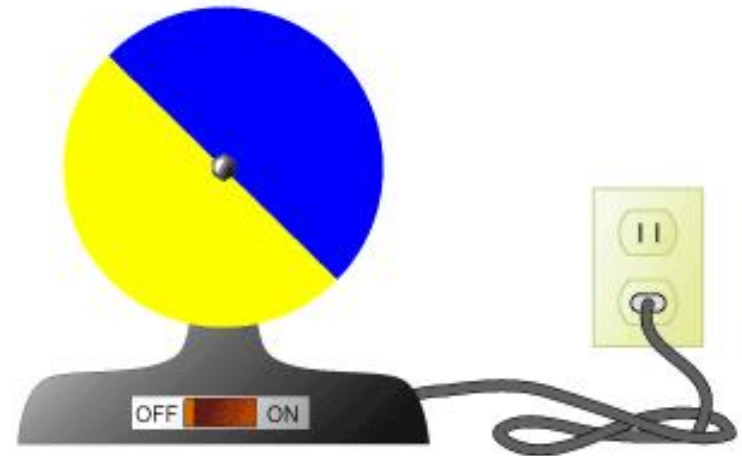
**FIGURE 6.2** Wavelengths comprising the visible range of the electromagnetic spectrum. (Courtesy of the General Electric Co., Lamp Business Division.)

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review



**Primary colors of light**



**Color wheel**

Rotate slowly



# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

$$C = X[R] + Y[G] + Z[B]$$

Eg:

$$\text{Red} = 1[R] + 0[G] + 0[B]$$

$$\text{Yellow} = 1[R] + 1[G] + 0[B]$$

$$\text{Cyan} = 0[R] + 1[G] + 1[B]$$

$$\text{Magenta} = 1[R] + 0[G] + 1[B]$$

$$\text{Black} = 0[R] + 0[G] + 0[B]$$

$$\text{White} = 1[R] + 1[G] + 1[B]$$

Hue 色调

Saturation 饱和度

Intensity 辉度

Chroma 色度

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

If the tristimulus values are denoted, X,Y and Z, then the trichromatic coefficients are defined as:

$$x = \frac{X}{X + Y + Z} \quad (2.3.1)$$

$$y = \frac{Y}{X + Y + Z} \quad (2.3.2)$$

$$z = \frac{Z}{X + Y + Z} \quad (2.3.3)$$

$$x + y + z = 1 \quad (2.3.4)$$

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

### Color Fundament: CIE chromaticity diagram

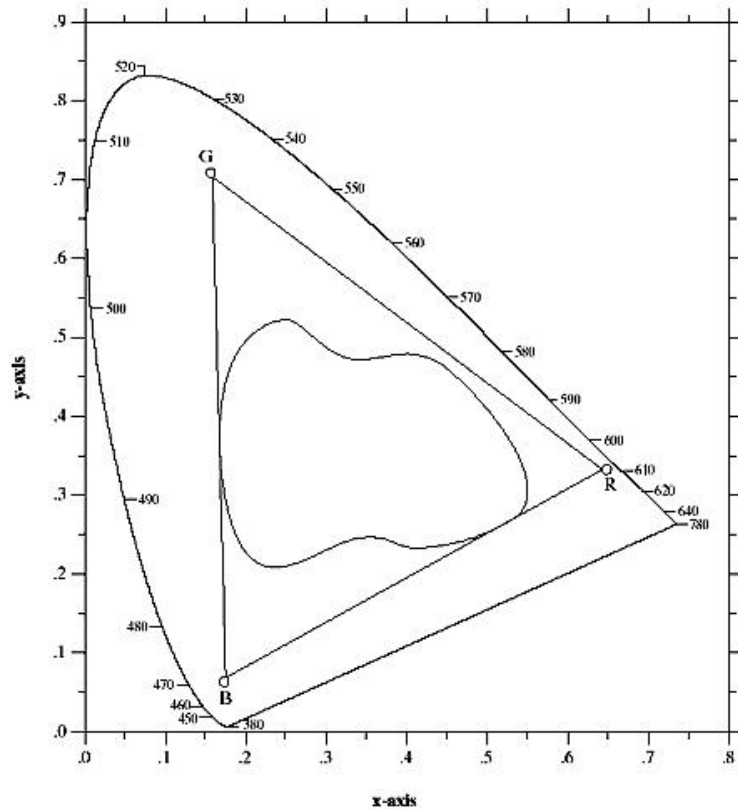
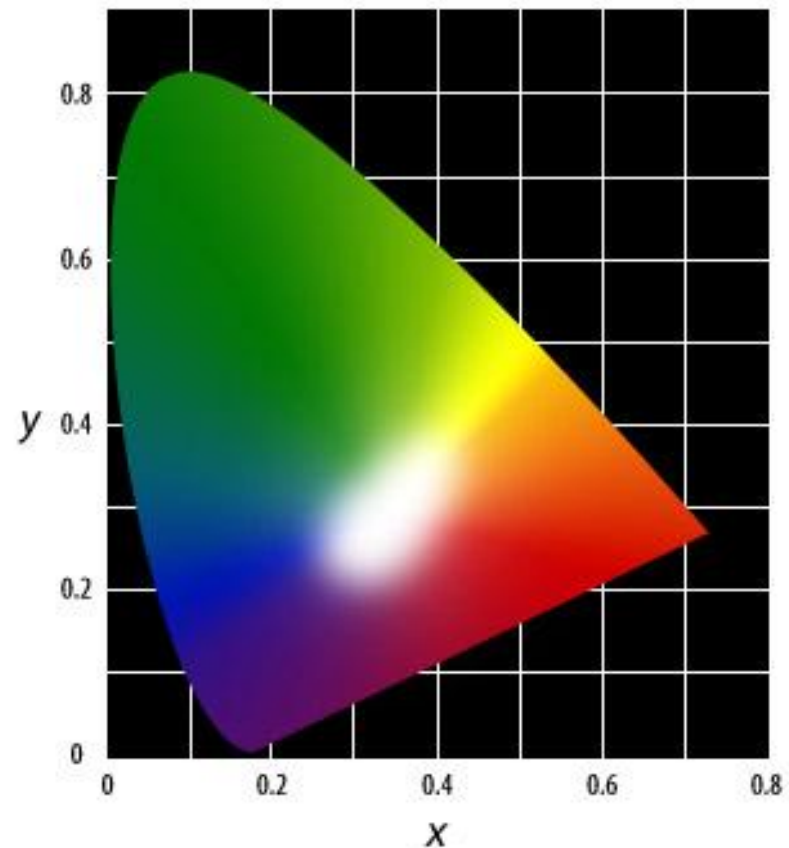


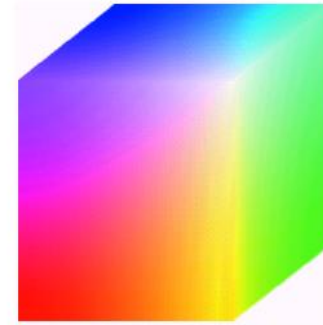
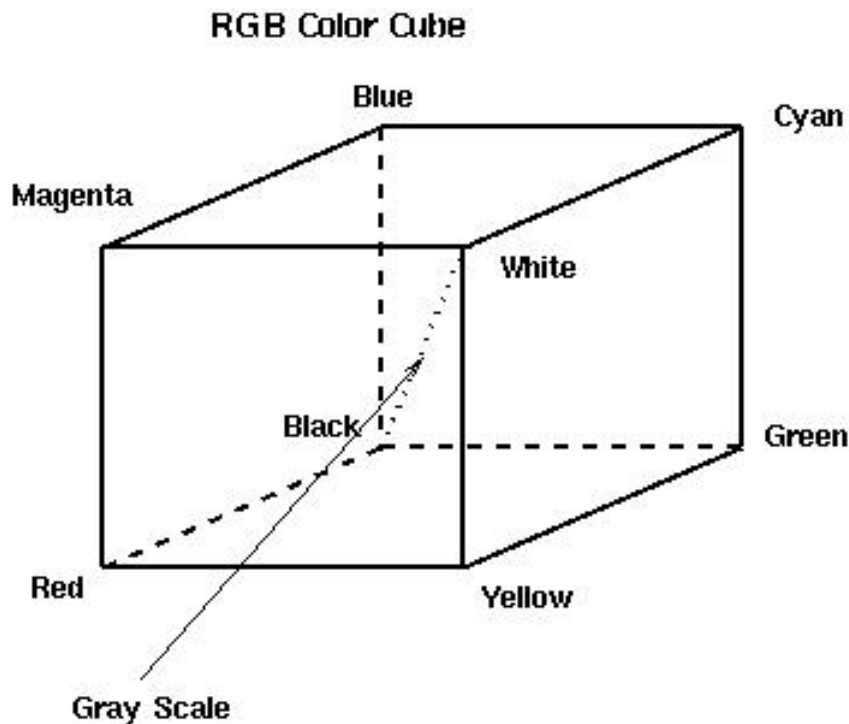
FIGURE 6.6 Typical color gamut of color monitors (triangle) and color printing devices (irregular region).



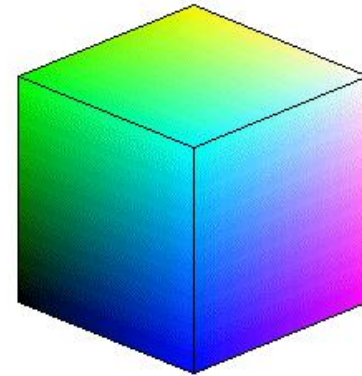
# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

### Color Model: RGB model



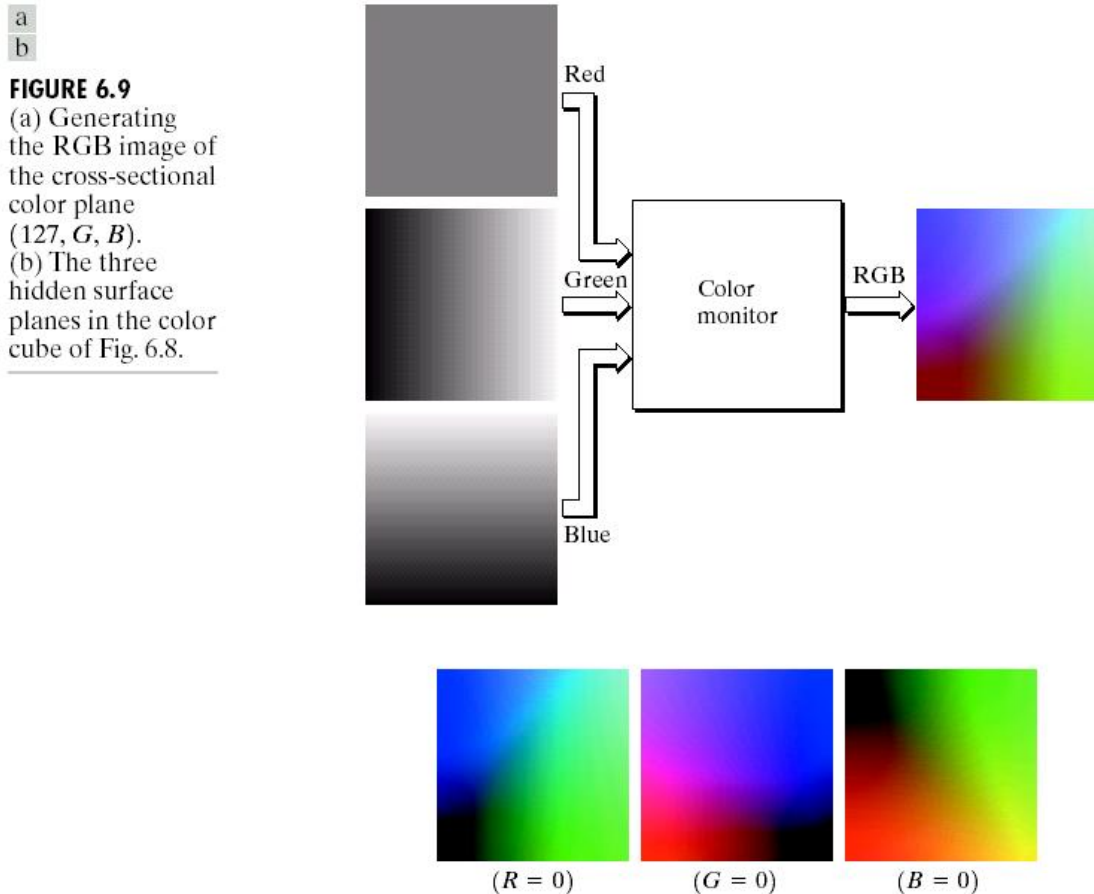
6.8 RGB 24-bit color cube.



# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

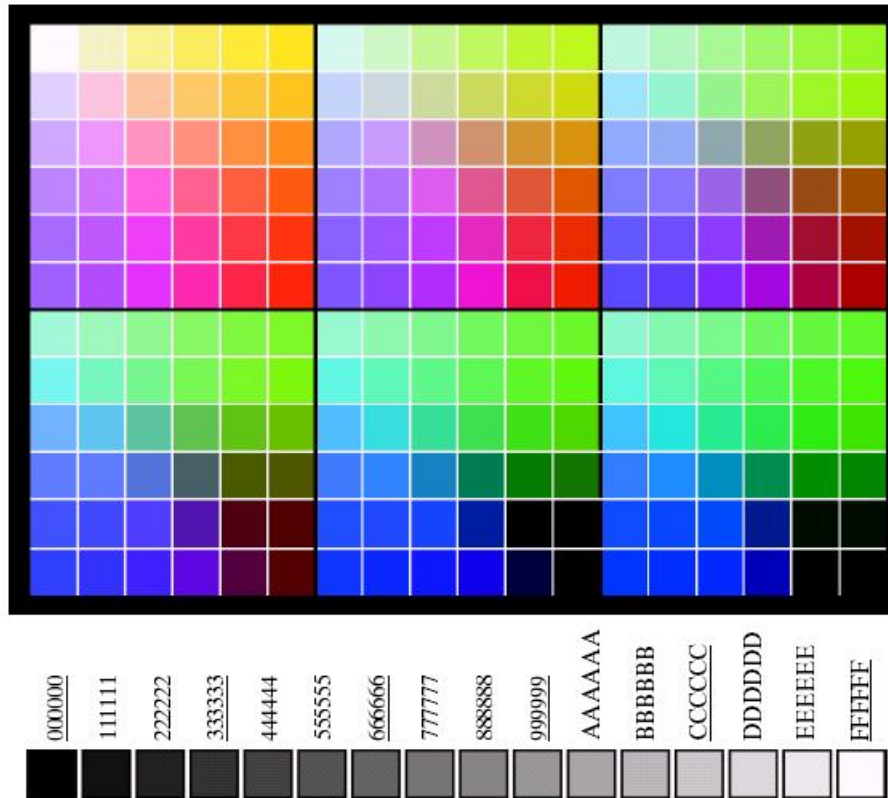
### Color Model: RGB model



# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

### Color Model: RGB model



a  
b

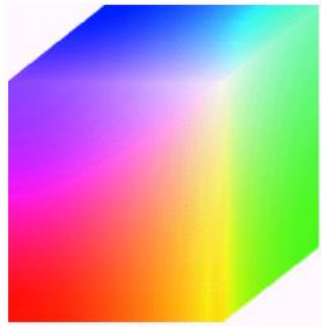
**FIGURE 6.10**

(a) The 216 safe RGB colors.  
(b) All the grays in the 256-color RGB system (grays that are part of the safe color group are shown underlined).

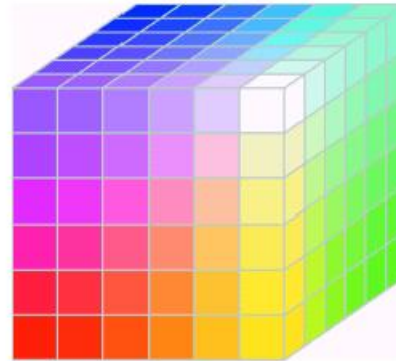
# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

Color Model: safe colors of RGB model



**FIGURE 6.8** RGB 24-bit color cube.



**FIGURE 6.11** The RGB safe-color cube.

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

Color Model: CMY model

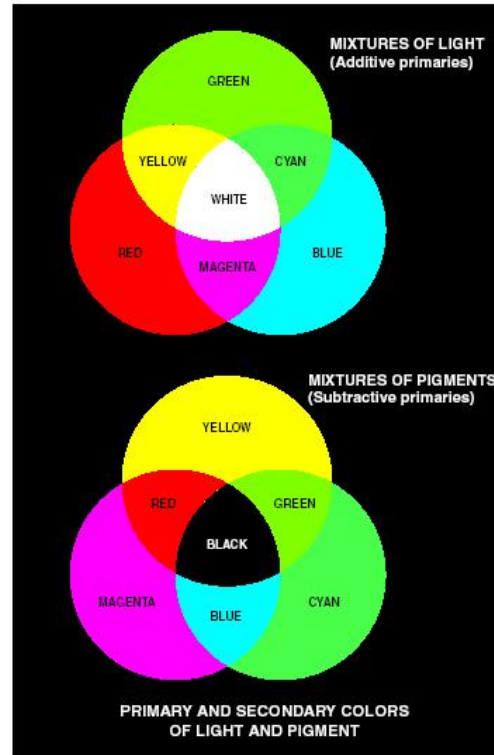
$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

### Color Model: CMY model



a  
b

**FIGURE 6.4** Primary and secondary colors of light and pigments. (Courtesy of the General Electric Co., Lamp Business Division.)

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

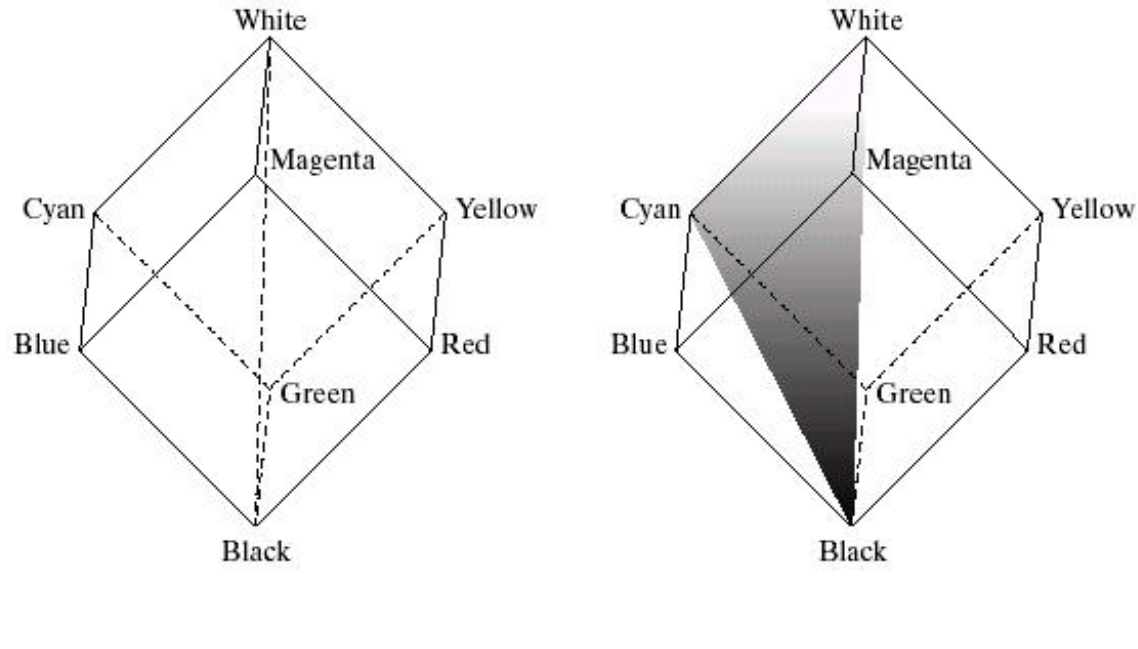
### Color Model: HSI model

- **Hue:** associated with the dominant wavelength in a mixture of light waves
- **Saturation:** refer to the relative purity or the amount of white light mixed with hue.
- **Intensity:** brightness

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

### Color Model: HSI model

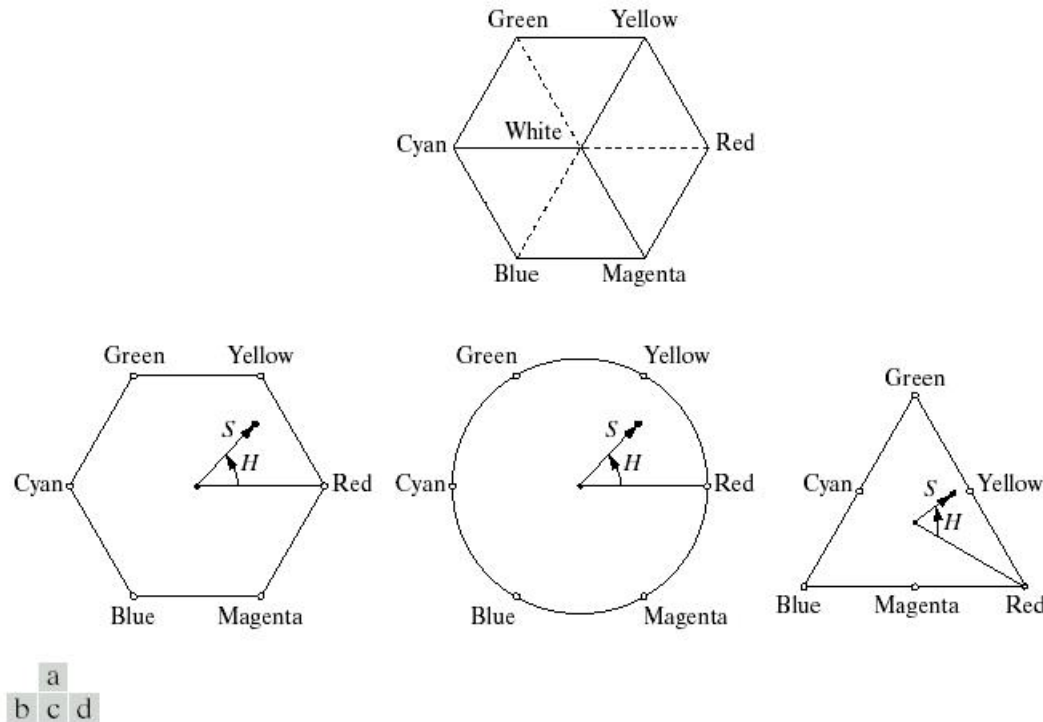


**FIGURE 6.12** Conceptual relationships between the RGB and HSI color models.

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

### Color Model: HSI model



**FIGURE 6.13** Hue and saturation in the HSI color model. The dot is an arbitrary color point. The angle from the red axis gives the hue, and the length of the vector is the saturation. The intensity of all colors in any of these planes is given by the position of the plane on the vertical intensity axis.

# 4.5 Color Image Enhancement

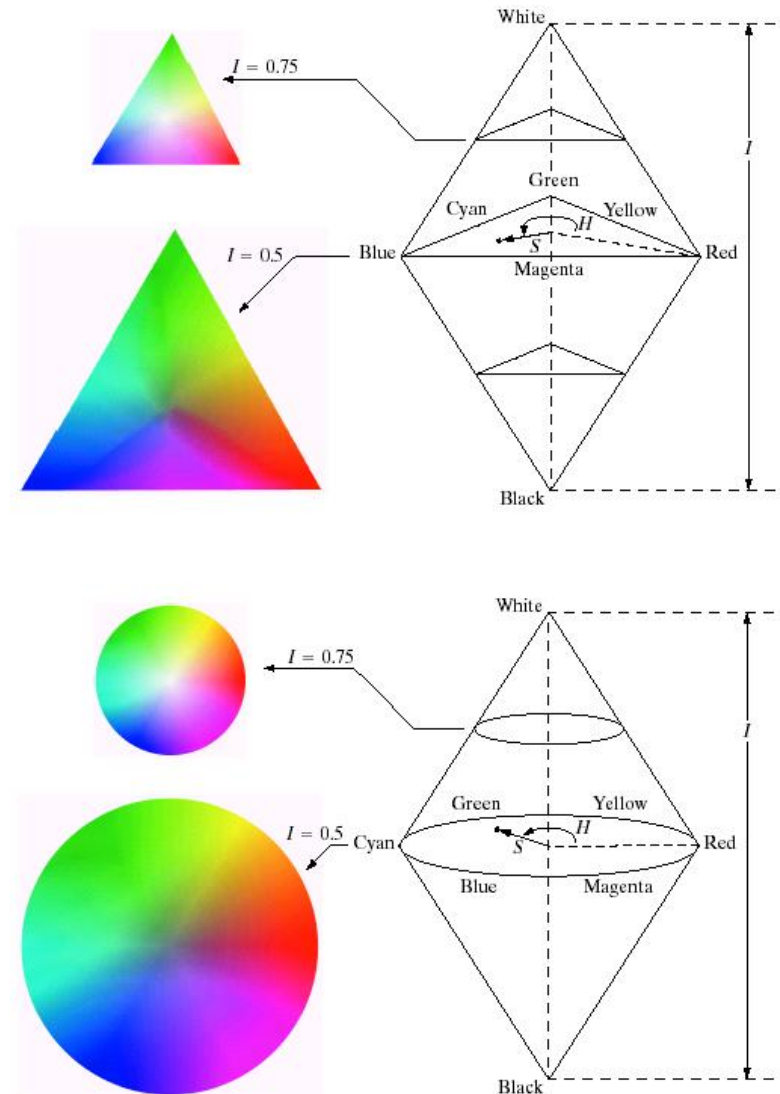
## 4.5.1 Introduction: review

### Color Model: HSI model

a  
b

**FIGURE 6.14** The HSI color model based on (a) triangular and (b) circular color planes. The triangles and circles are perpendicular to the vertical intensity axis.

| Color | H   | S | I   |
|-------|-----|---|-----|
| Red   | 0   | 1 | 0.5 |
| Green | 120 | 1 | 0.5 |
| Blue  | 240 | 1 | 0.5 |
| White | 0   | 0 | 1   |
| Black | 0   | 0 | 0   |



# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

Color Model: converting colors from RGB to HSI

$$I = \frac{1}{3}(R + G + B)$$

$$S = 1 - \frac{3}{(R + G + B)}[\min(R, G, B)]$$

$$\theta = \arccos \left\{ \frac{[(R - G) + (R - B)] / 2}{[(R - G)^2 + (R - B)(G - B)]^{1/2}} \right\}$$

$$H = \begin{cases} \theta & B \leq G \\ 360 - \theta & \text{others} \end{cases}$$

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

Color Model: converting colors from HSI to RGB

RG Sector:  $0^\circ \leq H < 120^\circ$

$$B = I(1 - S)$$

$$R = I \left[ 1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

$$G = 3I - (B + R)$$

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

Color Model: converting colors from HSI to RGB

GB Sector:  $120^\circ \leq H < 240^\circ$

$$R = I(1 - S)$$

$$G = I \left[ 1 + \frac{S \cos(H - 120^\circ)}{\cos(180^\circ - H)} \right]$$

$$B = 3I - (R + G)$$



# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

Color Model: converting colors from HSI to RGB

BR Sector:  $240^\circ \leq H < 360^\circ$

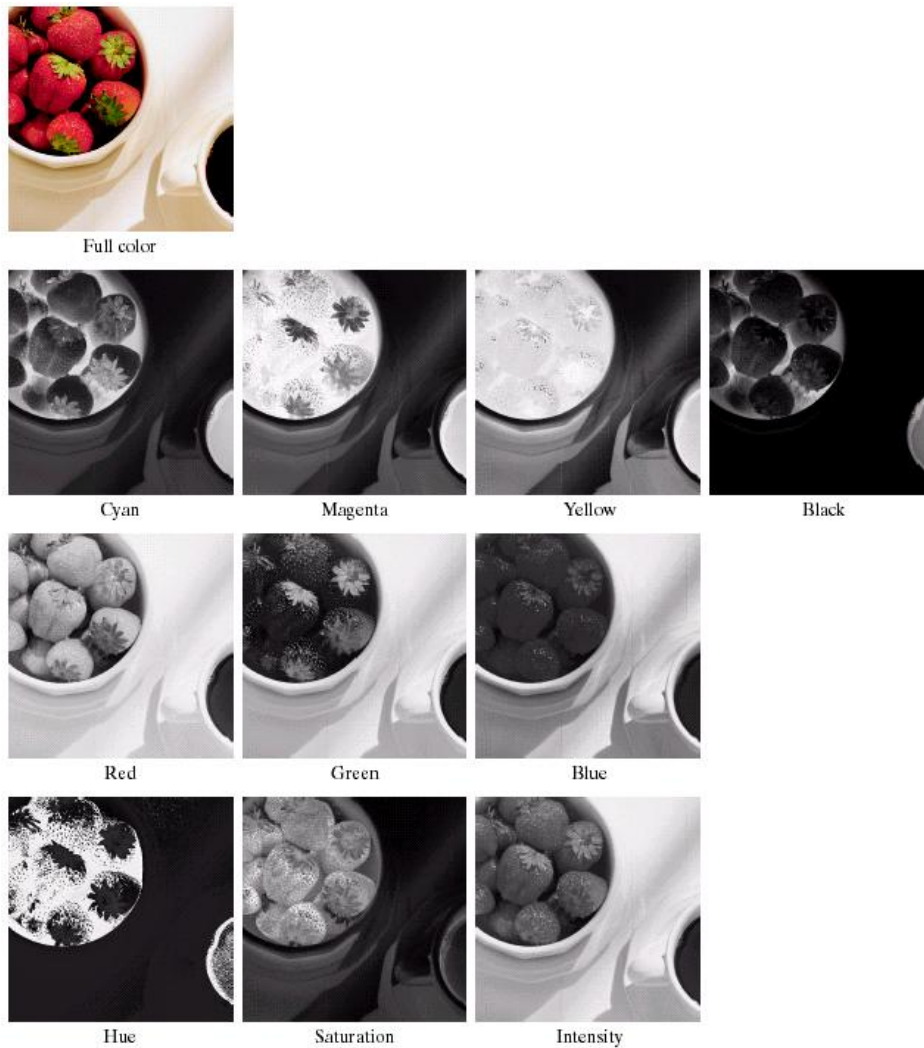
$$G = I(1 - S)$$

$$B = I \left[ 1 + \frac{S \cos(H - 240^\circ)}{\cos(300^\circ - H)} \right]$$

$$R = 3I - (G + B)$$

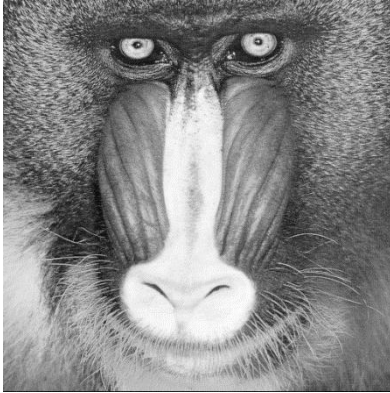
**FIGURE 6.30** A full-color image and its various color-space components. (Original image courtesy of Med-Data Interactive.)

---

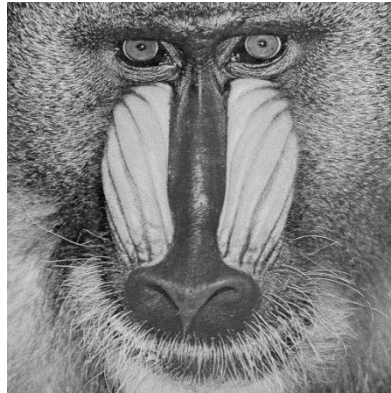


# 4.5 Color Image Enhancement

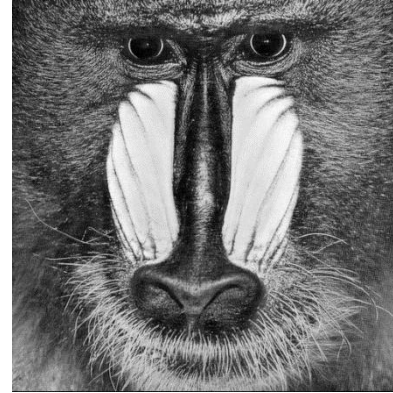
## 4.5.1 Introduction: review



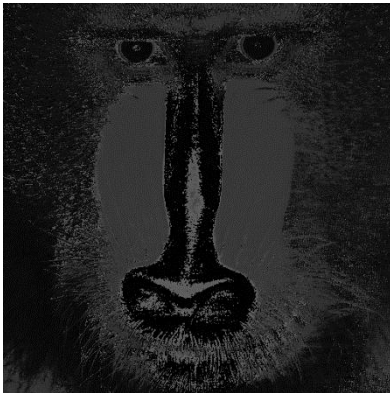
R



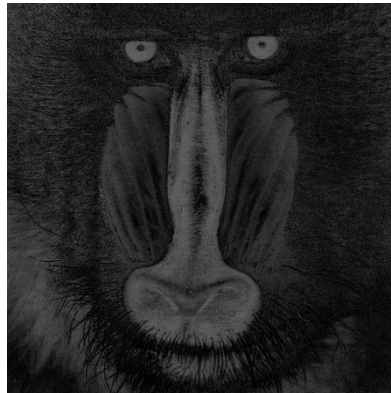
G



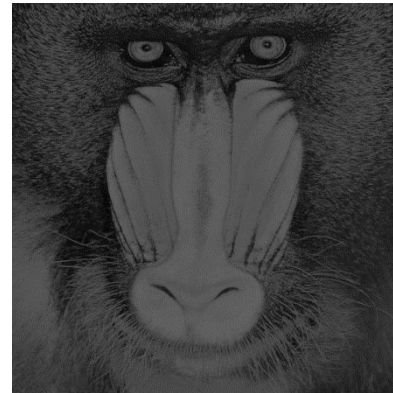
B



H



S

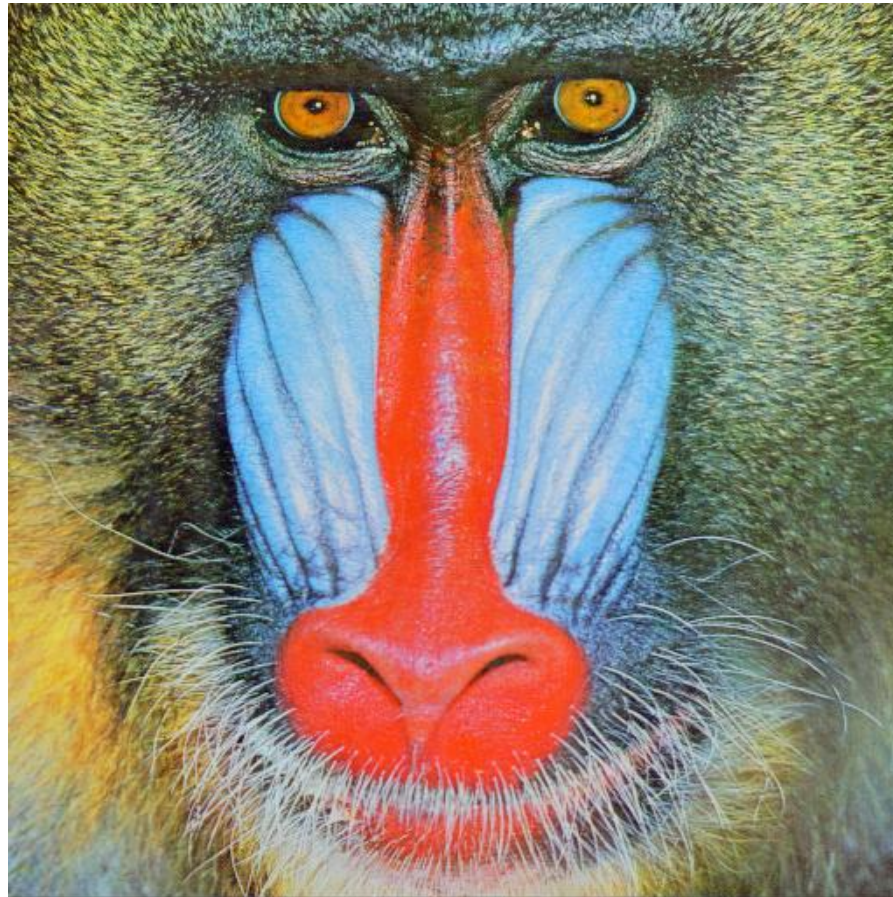


I

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

The answer is:

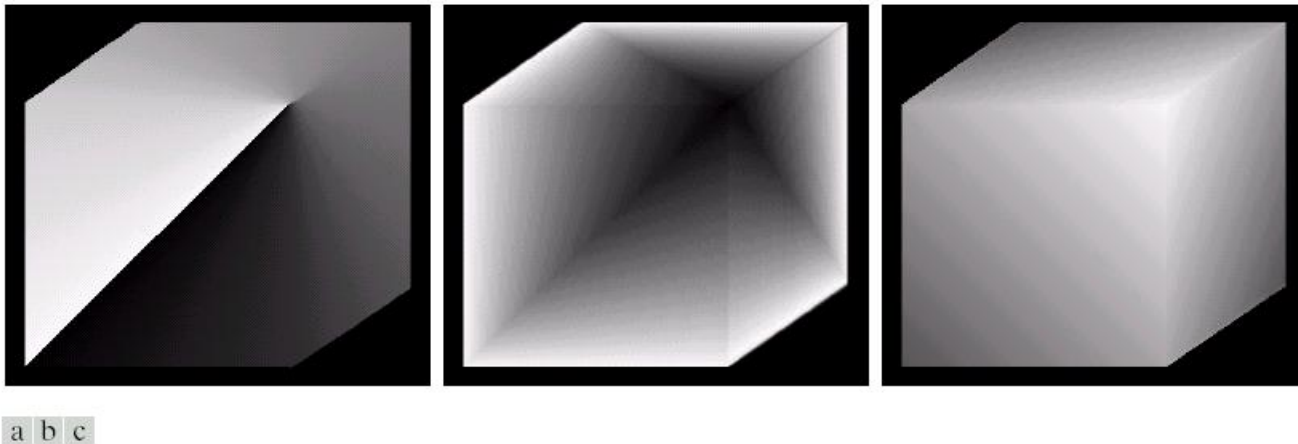




# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

Color Model: converting colors from HSI to RGB

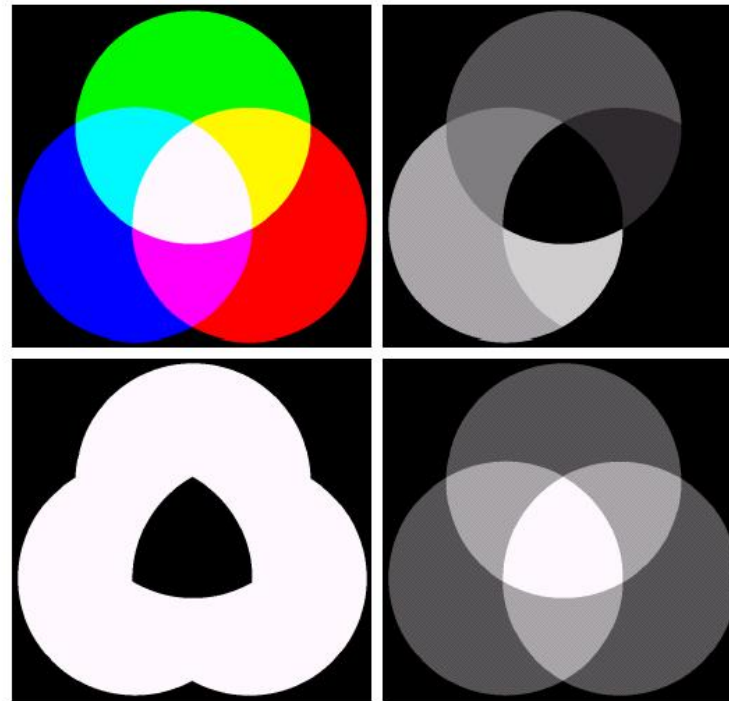


**FIGURE 6.15** HSI components of the image in Fig. 6.8. (a) Hue, (b) saturation, and (c) intensity images.

# 4.5 Color Image Enhancement

## 4.5.1 Introduction: review

Color Model: converting colors from HSI to RGB



a b  
c d

**FIGURE 6.16** (a) RGB image and the components of its corresponding HSI image: (b) hue, (c) saturation, and (d) intensity.



附加题： 从色彩定义的角度写出你对2015年安徽高考作文题的理解：

作文题如下：

阅读下面的材料，根据要求写一篇不少于800字的文章。

为了丰富中小学生的课余生活，让同学们领略科技的魅力，过一把尖端科技的瘾，中科院某研究所推出了公众开放日系列科普活动。活动期间，科研人员特地设计了一个有趣的实验，让同学们亲手操作扫描式电子显微镜，观察蝴蝶的翅膀。

通过这台可以看清纳米尺度物体三维结构的显微镜，同学们惊奇地发现：原本色彩斑斓的蝴蝶翅膀竟然失去了色彩，显现出奇妙的凹凸不平的结构。

原来，蝴蝶的翅膀本是无色的，只是因为具有特殊的微观结构，才会在光线的照射下呈现出缤纷的色彩.....

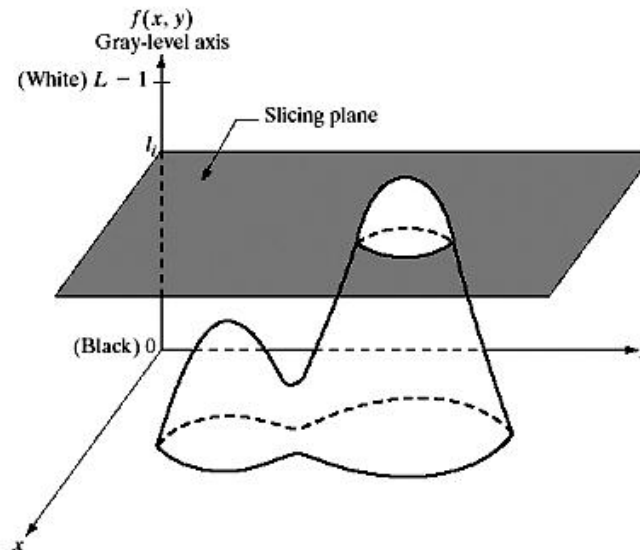
要求自选角度，确定立意，明确文体(诗歌除外)，自拟标题；

# 4.5 Color Image Enhancement

## 4.5.2 Pseudo color image processing : Intensity Slicing

formula

$$f(x, y) = c_k \quad \text{if } f(x, y) \in V_k$$

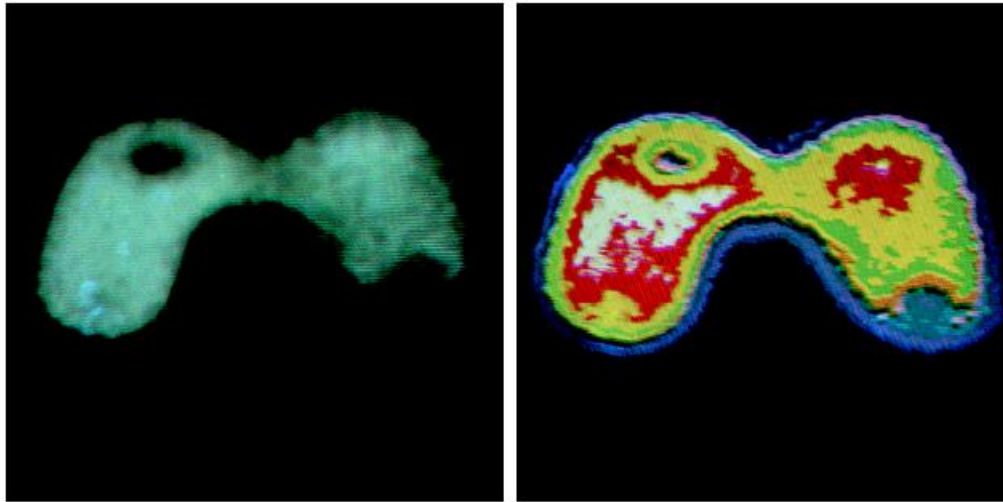


**FIGURE 6.18** Geometric interpretation of the intensity-slicing technique.



# 4.5 Color Image Enhancement

## 4.5.2 Pseudo color image processing : Intensity Slicing example



a b

**FIGURE 6.20** (a) Monochrome image of the Picker Thyroid Phantom. (b) Result of density slicing into eight colors. (Courtesy of Dr. J. L. Blankenship, Instrumentation and Controls Division, Oak Ridge National Laboratory.)

# 4.5 Color Image Enhancement

## 4.5.2 Pseudo color image processing : Intensity Slicing

example

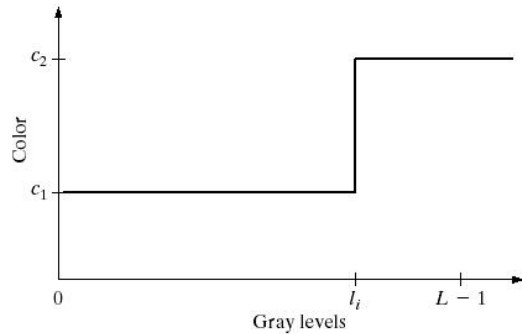
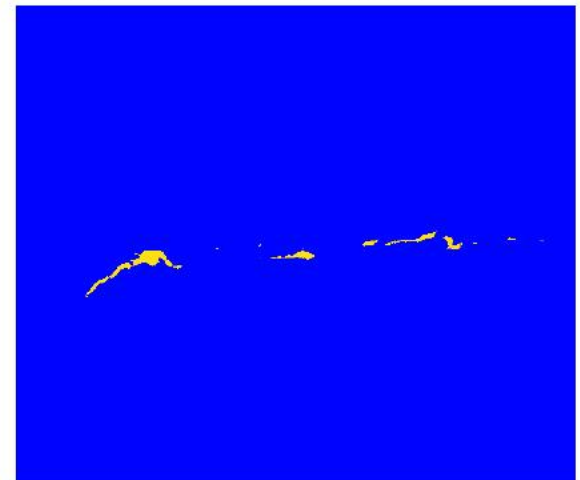
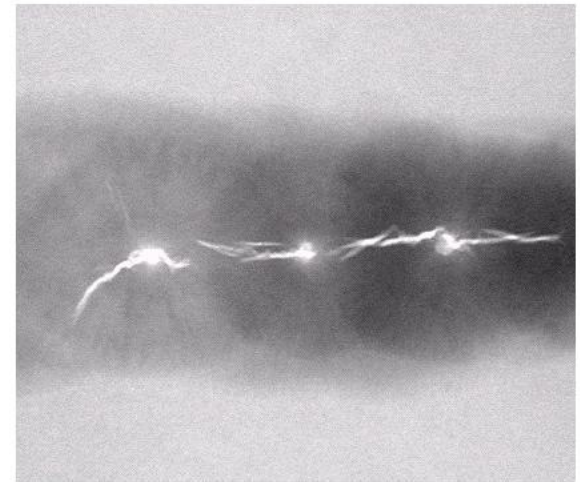


FIGURE 6.19 An alternative representation of the intensity-slicing technique.

a  
b

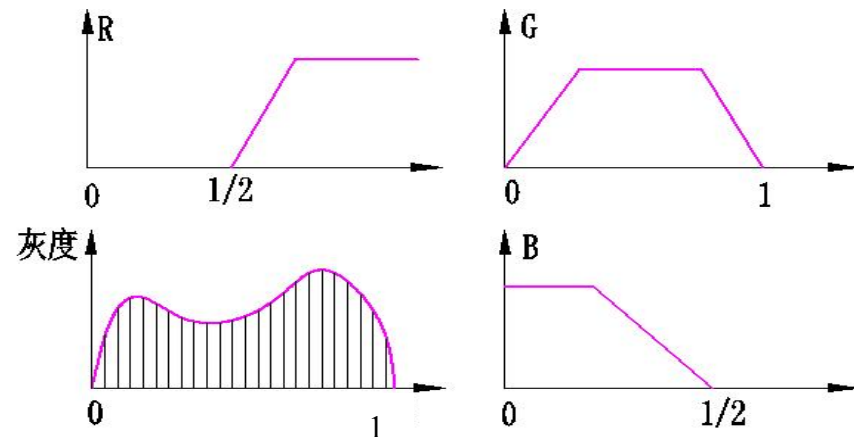
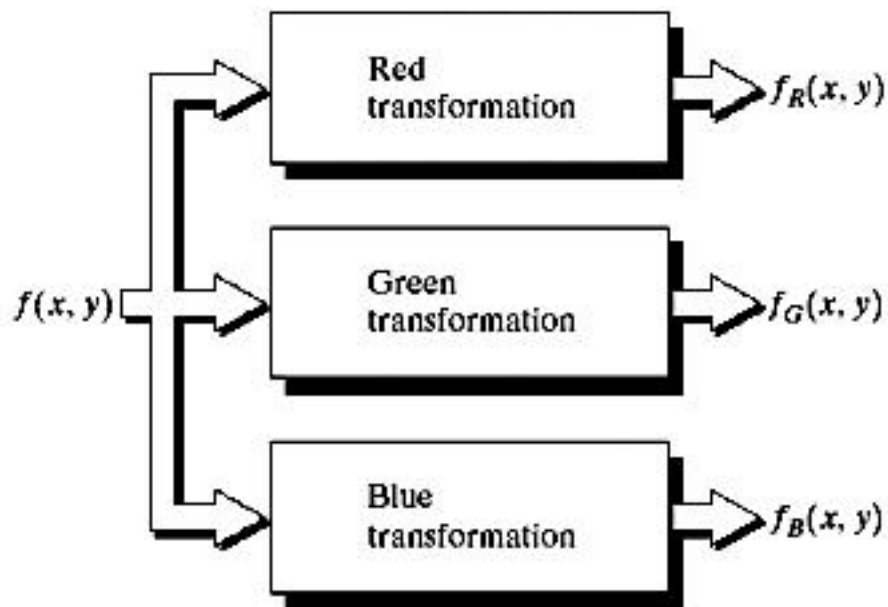
FIGURE 6.21  
(a) Monochrome X-ray image of a weld. (b) Result of color coding. (Original image courtesy of X-TEK Systems, Ltd.)



# 4.5 Color Image Enhancement

## 4.5.2 Pseudo color image processing : Gray level to color

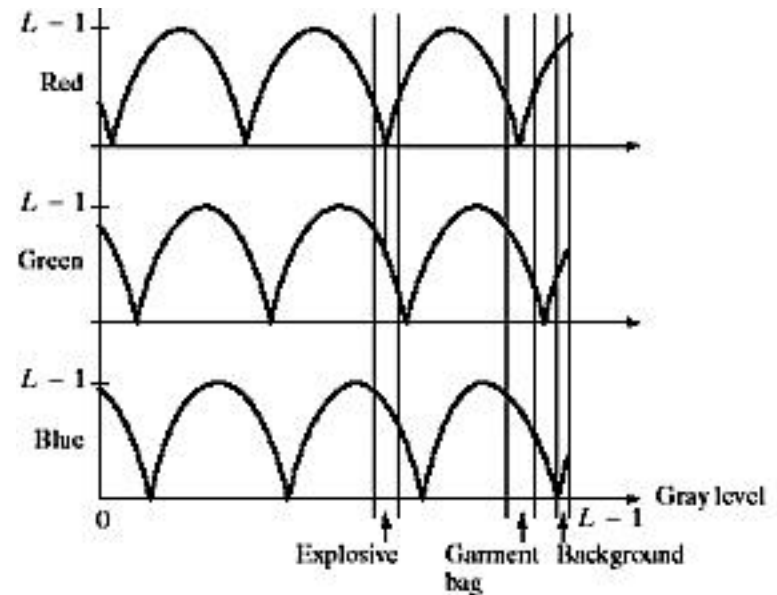
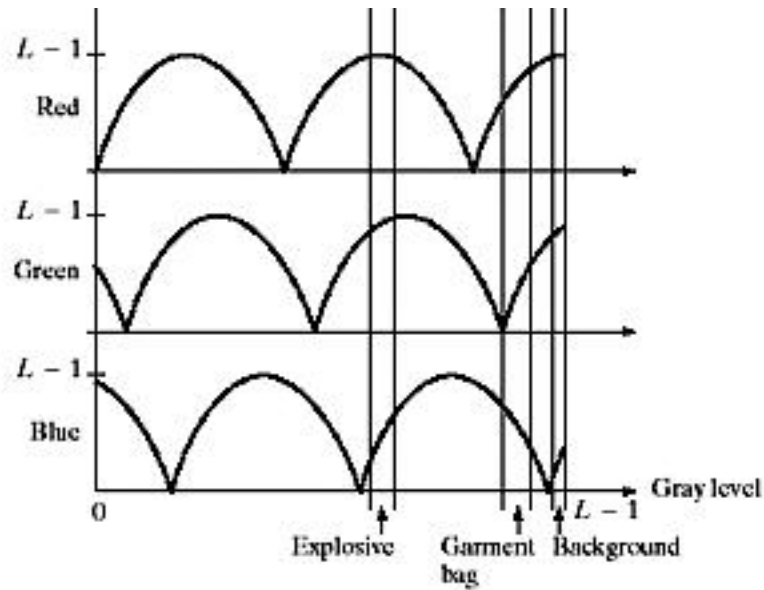
In general, we perform three independent transformations on the gray level of any input pixel



# 4.5 Color Image Enhancement

## 4.5.2 Pseudo color image processing : Gray level to color

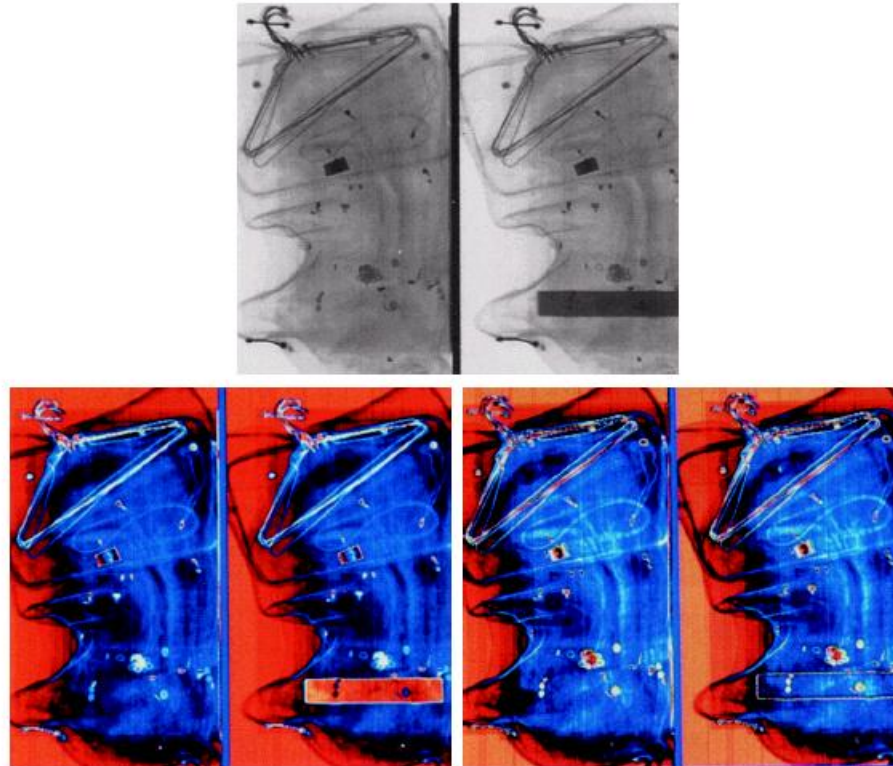
For example



transforms

# 4.5 Color Image Enhancement

## 4.5.2 Pseudo color image processing : Gray level to color

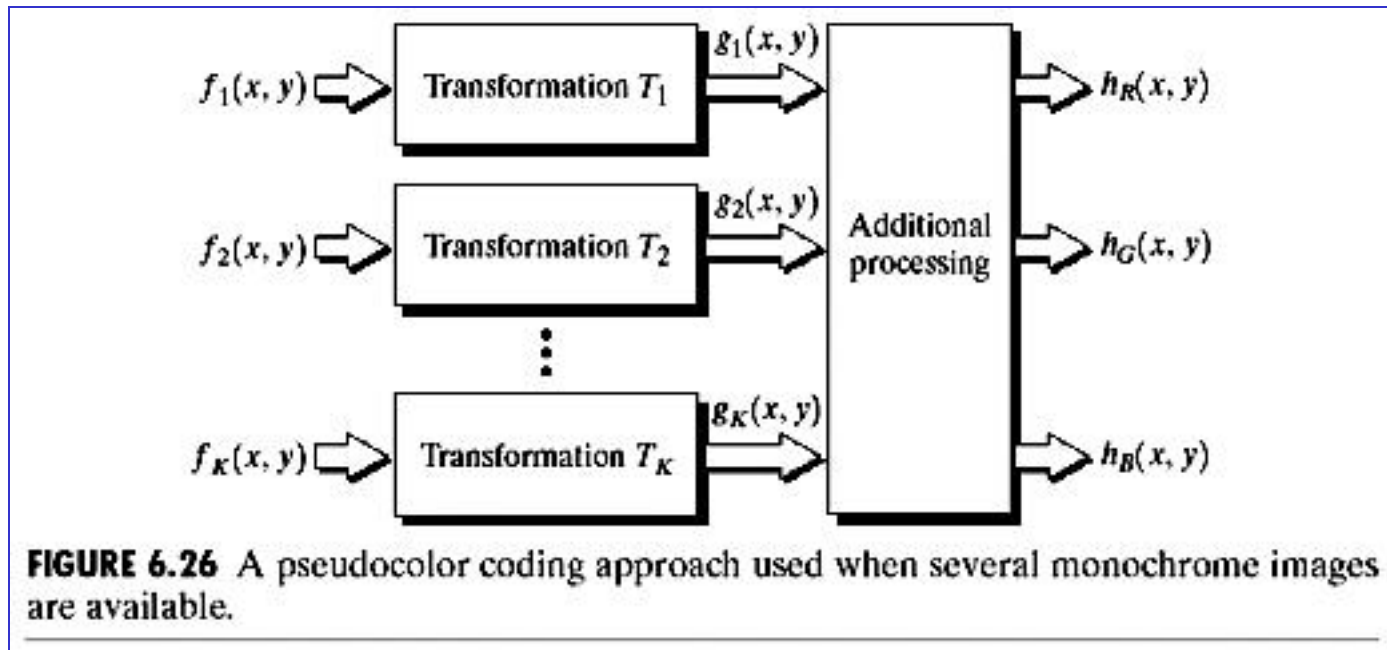


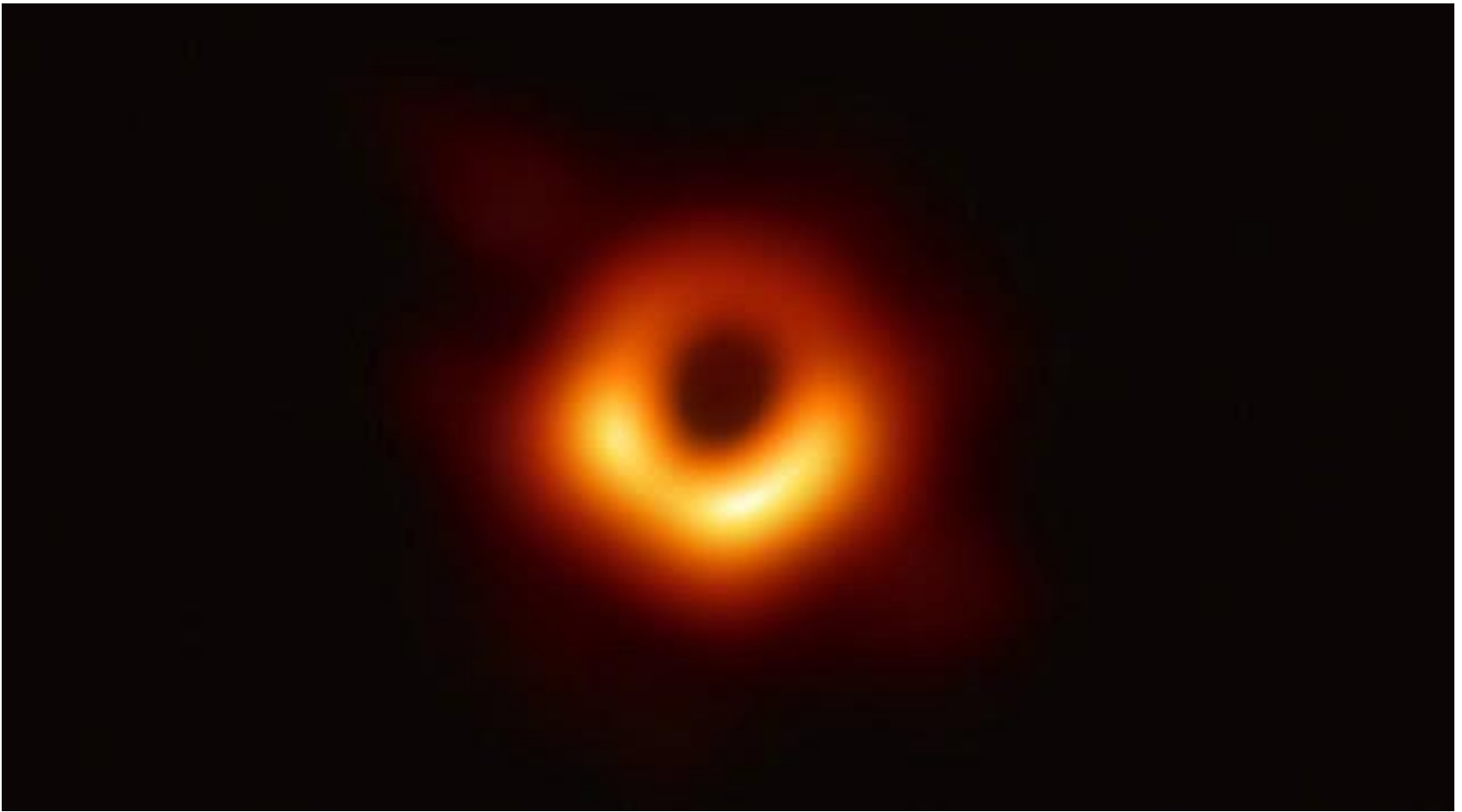
**FIGURE 6.24** Pseudocolor enhancement by using the gray-level to color transformations in Fig. 6.25. (Original image courtesy of Dr. Mike Hurwitz, Westinghouse.)

# 4.5 Color Image Enhancement

## 4.5.2 Pseudo color image processing : Gray level to color

Combine several monochrome images into a single color composite





拍黑洞的望远镜收集到的不是我们日常的可见光，而是一种波长更长的亚毫米波，

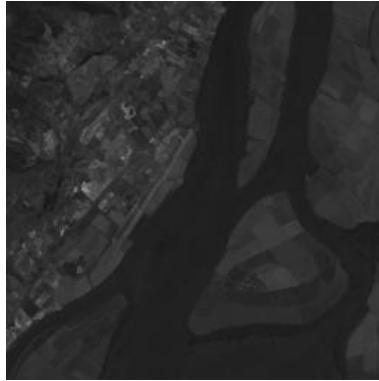


# 4.5 Color Image Enhancement

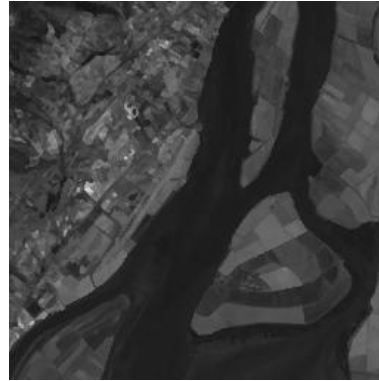
## 4.5.2 Pseudo color image processing : Gray level to color



band1



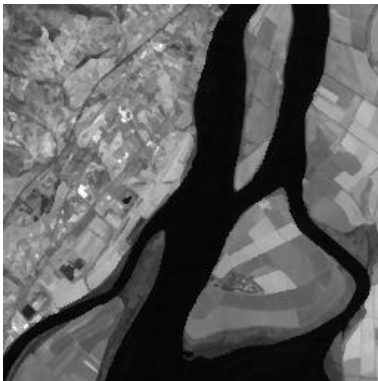
band2



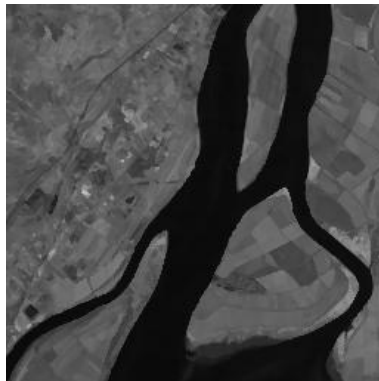
band3



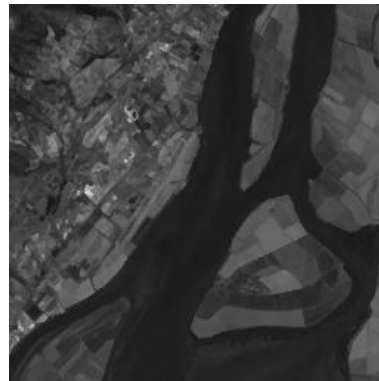
band1,2,3 false color



band5



band4



band3

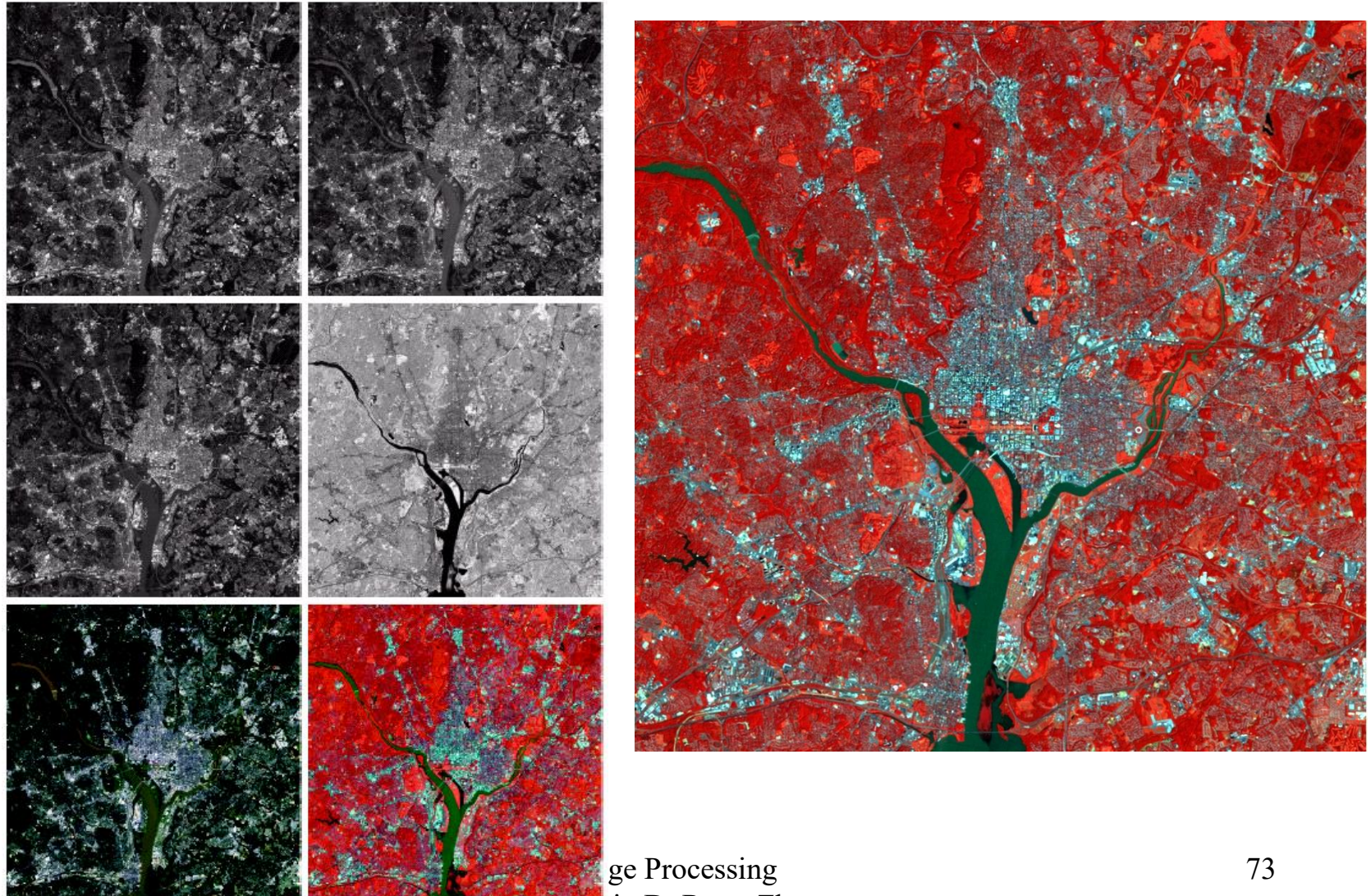


band5,4,3 false color



# 4.5 Color Image Enhancement

## 4.5.2 Pseudo color image processing : Gray level to color



# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : Fundamentals

A full color image can be expressed as

$$c(x, y) = \begin{bmatrix} c_R(x, y) \\ c_G(x, y) \\ c_B(x, y) \end{bmatrix} = \begin{bmatrix} R(x, y) \\ G(x, y) \\ B(x, y) \end{bmatrix}$$

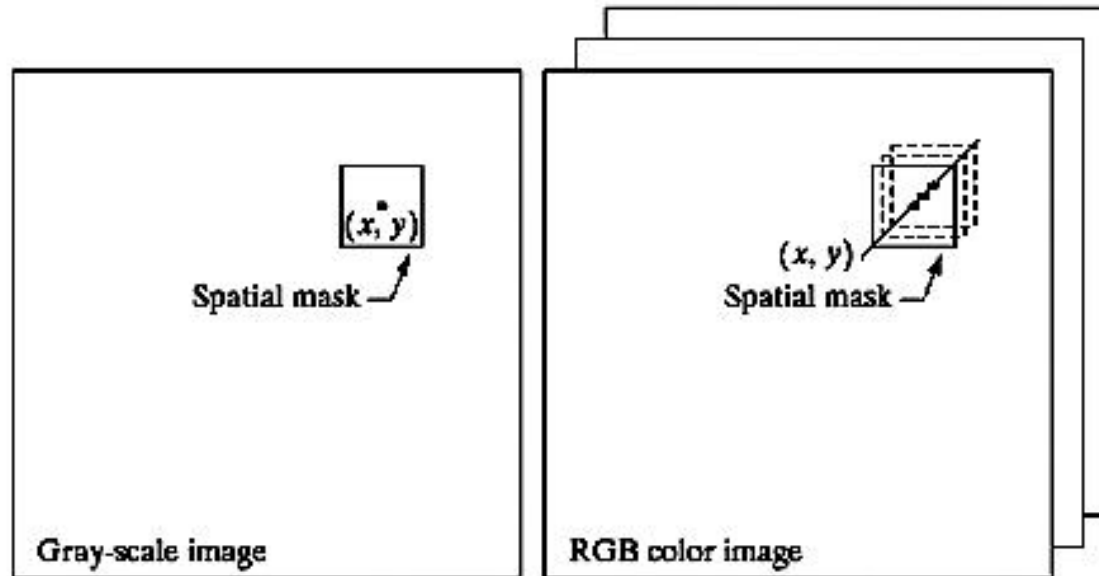
A pixel in color image interpreted as vector

$$c = \begin{bmatrix} c_R \\ c_G \\ c_B \end{bmatrix} = \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : Fundamentals

Mask operation in color image



# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : Color transformations

Adjusting the intensity of an image:

in RGB color space three components must be transformed:

$$s_i = kr_i \quad i = 1, 2, 3 \quad 0 < k < 1$$

in HSI color space only intensity components is modified:

$$s_3 = kr_3$$

$$s_1 = r_1$$

$$s_2 = r_2$$



# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : Color transformations

a b  
c d e

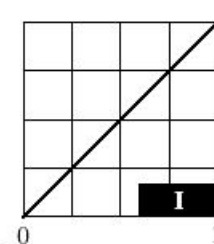
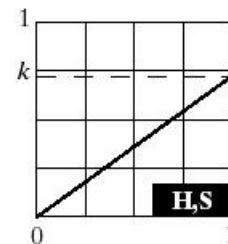
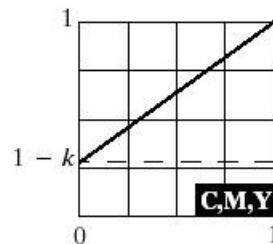
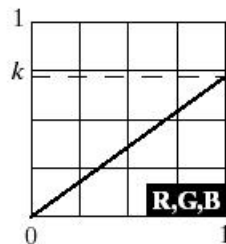
**FIGURE 6.31**

Adjusting the intensity of an image using color transformations.

(a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting  $k = 0.7$ ).

(c)–(e) The required RGB, CMY, and HSI transformation functions.

(Original image courtesy of MedData Interactive.)



I

HS

S

HI

# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : Color Complement

in RGB color space three components must be transformed:

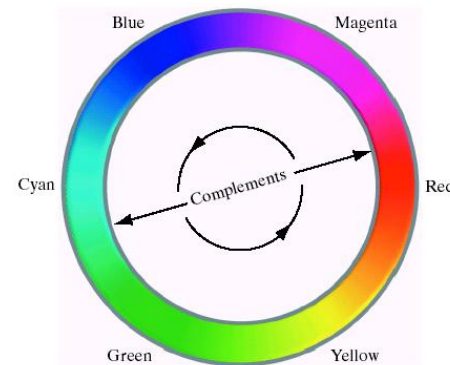
$$s_i = 1 - r_i \quad i = 1, 2, 3 \quad 0 < k < 1$$

in HSI color space only intensity components is modified:

$$s_1 = (r_1 + 0.5)$$

$$s_2 = r_2$$

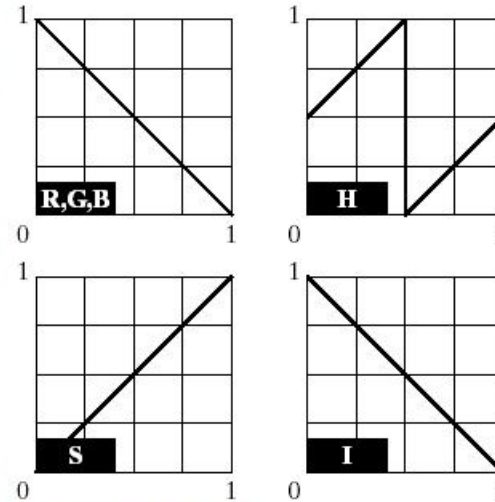
$$s_3 = 1 - r_3$$



**FIGURE 6.32**  
Complements on  
the color circle.

# 4.5 Color Image Enhancement

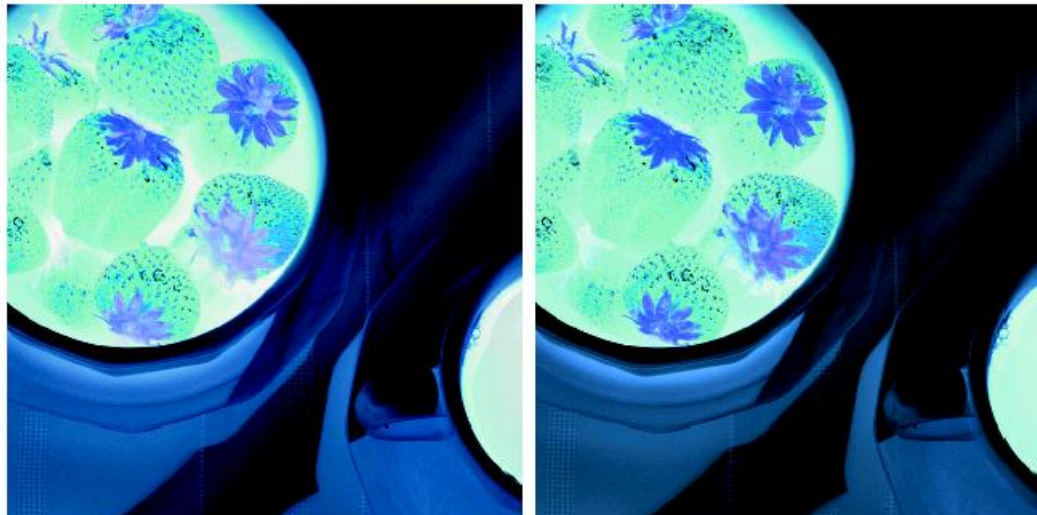
## 4.5.3 Full color enhancement : Color Complement



|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 6.33**

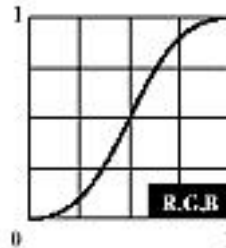
Color complement transformations. (a) Original image. (b) Complement transformation functions. (c) Complement of (a) based on the RGB mapping functions. (d) An approximation of the RGB complement using HSI transformations.



# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : color corrections

In RGB and CMY(K) space, map all three (or four) color components with the same transform function

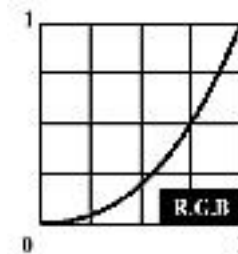




# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : color corrections

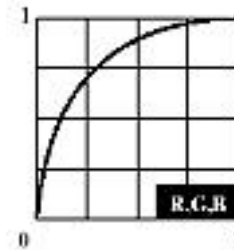
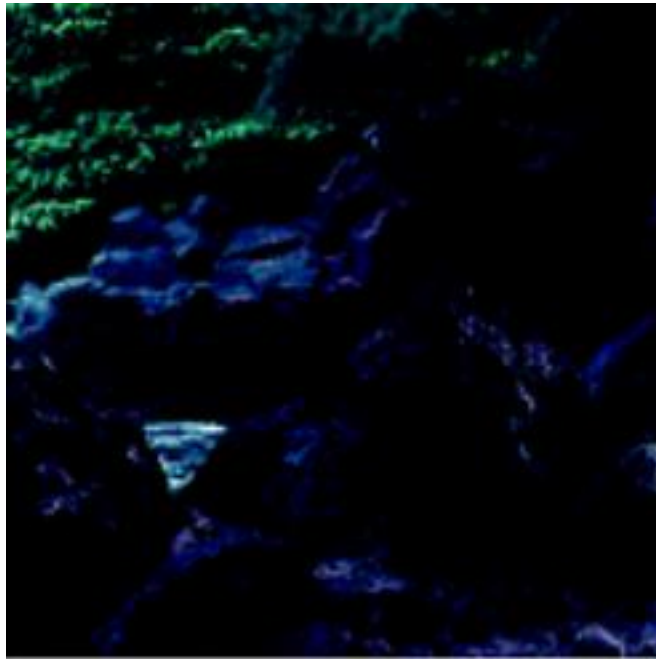
example



# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : color corrections

example



# 4.5 Color Image Enhancement

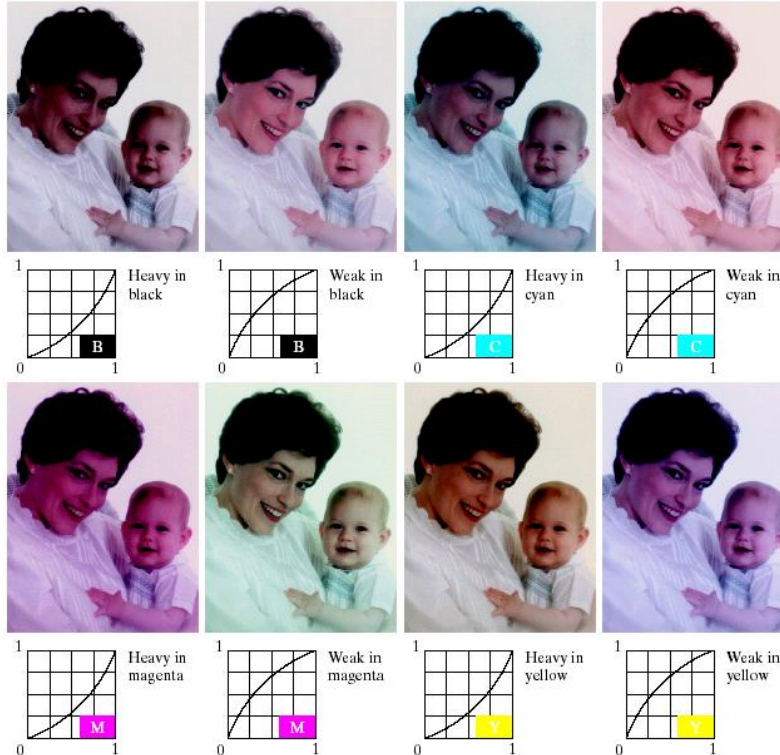
## 4.5.3 Full color enhancement : color corrections

example



Original/Corrected

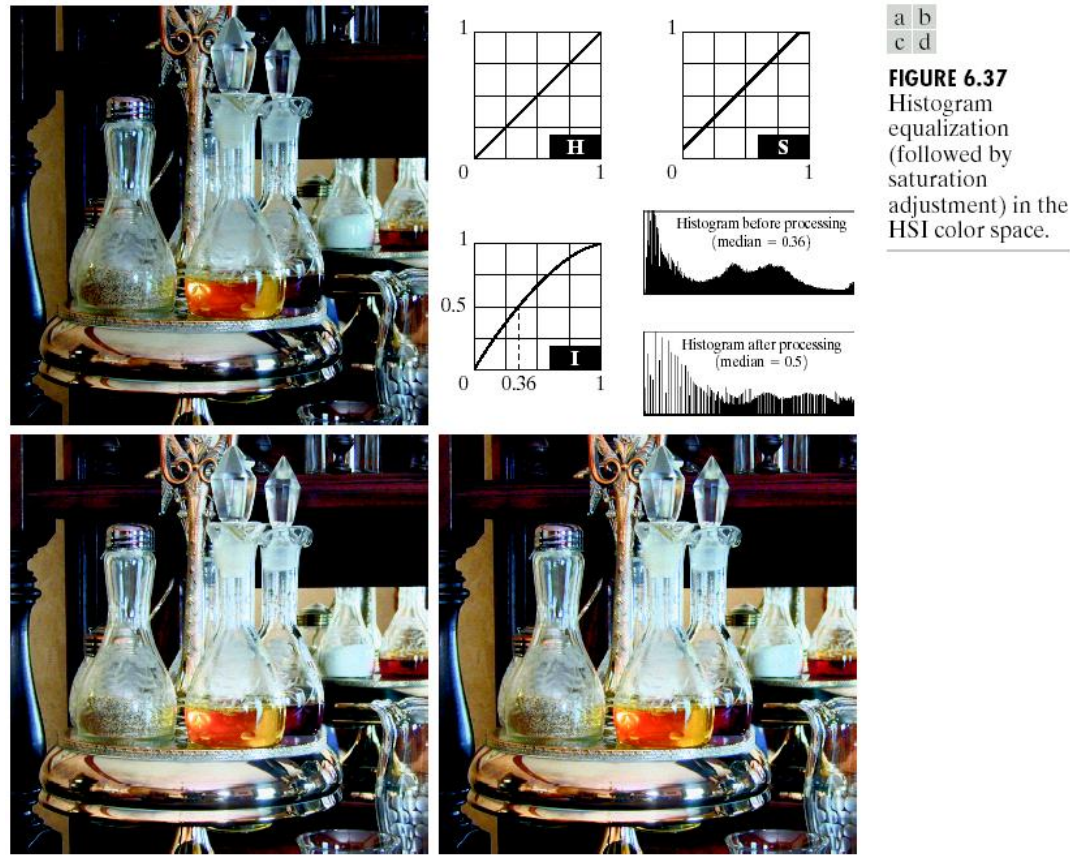
FIGURE 6.36 Color balancing corrections for CMYK color images.



# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : Histogram processing

In the HSI color space, only the intensity component is modified





# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : Histogram processing



原图

I分量均衡化

RGB分量均衡化

# 4.5 Color Image Enhancement

## 4.5.3 Full color enhancement : Smoothing and Sharpening



a b c

**FIGURE 6.40** Image smoothing with a  $5 \times 5$  averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.



a b c

**FIGURE 6.41** Image sharpening with the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the intensity component and converting to RGB. (c) Difference between the two results.



# 4.5 Color Image Enhancement

## 4.5. 6 Noise in color image :

### Noise in RGB channels

a b  
c d

**FIGURE 6.48**  
(a)–(c) Red, green, and blue component images corrupted by additive Gaussian noise of mean 0 and variance 800. (d) Resulting RGB image. [Compare (d) with Fig. 6.46(a).]



# 4.5 Color Image Enhancement

## 4.5. 6 Noise in color image :

Noise in HSI channels



a b c

**FIGURE 6.49** HSI components of the noisy color image in Fig. 6.48(d). (a) Hue. (b) Saturation. (c) Intensity.



## Questions?

- 1、 在一条自动装配线上，有三类形状相同的工件，为了检测方便，将工件用不同颜色标注，现只有1个单色摄象机，请提出一种用这个摄象机检测3种颜色的方法。
- 2、 设有一个能输出RGB模拟信号的彩色摄象机，一个能将这 些模拟信号转化为以  $(1/30)$  s的视频速度输出RGB或HIS图象的数字化器，3块能以视频速度接受图象的帧缓存卡，以及一个能以视频速度计算直方图的硬件。所有这些都可以与1台微机组组合在一起，现要解决以下问题：生产线上有一系列形状相同但颜色不同的工件，他们按红、黄、绿、蓝的次序排列。请借助以上硬件设计一个图象处理软件系统将工件的颜色检测出来。这里假设工件移动速度相当慢，所以可忽略由此产生的图象模糊问题，请画出系统的流程图并对每个模块及所选处理技术进行讨论。

## Homework:

- 1、为什么一般情况下对离散图象的直方图均衡并不能产生完全平坦的直方图？
- 2、设已用直方图均衡化技术对一幅图象进行了增强，试证明再用这个方法对所得结果增强并不会改变其结果
- 3、讨论用于空间滤波的平滑滤波器和锐化滤波器的相同点、不同点以及联系。
- 4、有一种常用的图象增强技术是将高频增强和直方图均衡化结合起来以达到使边缘锐化的反差增强效果，以上两个操作的先后次序对增强结果有影响吗？为什么？
- 5、编程实现对lena.bmp分别加入高斯噪声和椒盐噪声，再进行局域平均和中值滤波。

The End